Optimizing Methods for Feynman Integral Reduction with Kira

Abstract

Feynman integral reduction expresses the loop integrals in a particle-physics calculation through a smaller set of master integrals. We studied how machine learning and exact analytic identities can reduce the time required to complete this calculation with Kira. We first constructed 2.30 million targets with known master expansions and recorded seed and elimination data. We trained seed and row models, tested their algebraic sufficiency, and then evaluated their cost inside Kira. We then developed equation selectors and models of complete reduction time, combined them with selective seed enlargement and master-basis conversion, and added shared calculations and analytic reductions. Model gains depend on the workload: a shared structural model reduces time by 4.9% against a matched fixed strategy on 43 completed native requests, while the row model gives no reliable retained gain. Exact recurrences give a much larger but narrower result. For a specified coupled-vacuum family at one through twelve loops, a direct comparison reduces total time from 1,027.781 to 1.922 seconds. Later direct formulas and coefficient reconstruction reduce a separate 75-request workload from 81.827 to 49.852 seconds. The final 93-request update gives a further, separately measured 8.8% reduction against its preceding application. We explain the data, models, mathematical identities, controls, and unsuccessful approaches that led to these results. Restricted analytic families extend through thirteen loops; general high-loop scattering remains outside the demonstrated timing gains.

1 Introduction

Predictions for particle scattering and decay involve amplitudes, whose squared magnitudes enter measurable probabilities and rates. In perturbation theory, an amplitude is calculated as an expansion in the interaction strength. Higher-order terms contain integrations over internal momenta that are not fixed by the external particles. These integrations give Feynman integrals, and the number of independent internal momenta is the number of loops [1]. Quantum chromodynamics, the theory of the strong interaction, requires these integrals in higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes [1].

Many integrals share the same denominator factors and differ only in their powers or numerators. Integration-by-parts (IBP) identities relate such integrals, so the calculation can express them through master integrals: a set of integrals that remains after the chosen reduction relations have been used [2, 3]. Kira automates the construction and solution of these relations [4]. We call this native reduction when Kira generates and solves the IBP equations. Finding the coefficients of the masters is the reduction problem studied here. Evaluating the remaining masters is a further step, although some of the analytic formulas used below also provide their values.

The question is how to complete the requested reduction with less elapsed time. Kira already has methods for restricting equation generation and removing unnecessary relations [5]. A further restriction can save work, but an insufficient equation system requires recovery. Our study therefore compares complete calculations, including the choice of method and any repeated attempt. Machine-assisted seed search provides a related approach to reducing the equation system [6]; analytic reduction provides another route when the denominator factors satisfy suitable conditions [7, 8].

We developed and tested a combined reduction procedure. We constructed targets with known answers, trained seed and row models, integrated them with Kira, and then trained cost models and derived exact reductions to avoid repeated algebra. We ask both whether learning helps and whether the mathematical structure permits a cheaper calculation. A positive result must specify its physical domain and its timed control; a failed attempt remains part of the cost.

Section 2 introduces the integrals and IBP reduction. Section 3 explains the training data and the original models. Section 4 follows the development of the combined method, and Section 5 explains how we compare it fairly. Section 6 brings together the measured gains, including direct comparisons with original baselines. Section 7 relates each measured gain to its integral structure and physical use. The discussion compares the methods and states the physical limits, while the appendices give derivations, input definitions, and the detailed final incremental test. The companion notes retain the full experiment history.

2 From a loop integral to a master-integral reduction

We first define the notation used in this paper. This makes it possible to state both the reduction problem and the conditions under which an exact shortcut applies.

2.1 What the integrals contain

We use the normalized integral

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{j=1}^{N} D_j^{-a_j}, \qquad D_j = Q_j^2 - m_j^2 + i0, \qquad d = 4 - 2\epsilon. \tag{1}$$

Here L is the number of loops, ℓ_h is an integration momentum, and N is the number of denominator coordinates. Each Q_j is a specified linear combination of loop momenta and external momenta, and m_j is a mass. The vector $\boldsymbol{x}$ contains squared masses and scalar products of external momenta. The symbol i is the imaginary unit, and $i0$ means an infinitesimal positive imaginary part that specifies the Feynman pole prescription. An external invariant is a scalar product that does not change between inertial reference frames. For a physical internal line, $1/D_j$ is its scalar propagator and D_j is its inverse. The normalization $i\pi^{d/2}$ is conventional. Analytic continuation extends an integral formula from a region where the integral converges to other parameter values. Dimensional regularization uses this continuation in d; its poles describe divergences as the regulator ϵ approaches zero [1].

The integer vector $\boldsymbol{a} = (a_1, \ldots, a_N)$ identifies an integral. A positive a_j places $D_j^{a_j}$ in the denominator, $a_j = 0$ removes that factor, and a negative a_j places a power of D_j in the numerator. An *integral family* fixes all Q_j, masses, and external scalar products while allowing these indices to vary. Auxiliary coordinates represent numerator scalar products that the physical denominators do not span [4]. Thus a family specifies the mathematical problem before any model makes a choice.

For example, a one-loop two-point integral, commonly called a bubble, has the form

$$B_{ab}(p^2; x, y) = \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{(\ell^2 - x + i0)^a [(\ell - p)^2 - y + i0]^b}. \tag{2}$$

This equation defines x and y as squared masses, p as the external momentum, and a, b as denominator powers. The integrals B_{11} and B_{21} belong to the same family and have different powers of the first denominator. More complicated families introduce additional loop momenta and denominator factors, using the same notation.

The chosen set of masters is called a master basis. A *target* is an integral whose reduction is requested. A *sector* groups integrals with the same positive-index positions, and a sub-sector

removes one or more of those positions [4]. A request in our experiments contains one family, its physical parameters, and one or more targets. This distinction matters because a single timing measurement can cover several targets.

2.2 How integration by parts relates the targets

Integration by parts supplies an equation by integrating a total derivative. A seed is an integer index vector at which we generate these identities. For seed indices $\boldsymbol{\sigma}$ and a loop or external momentum v, the identity is [2]

$$0 = \int \prod_h \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_j D_j^{-\sigma_j} \right) = \sum_u A_{\boldsymbol{\sigma},o;u}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(u)}}. \tag{3}$$

The repeated component index μ is summed, k selects the differentiated loop momentum, and $o = (k, v)$ labels the derivative operation. Expanding the derivative changes denominator powers, giving the coefficients $A_{\boldsymbol{\sigma},o;u}$, where u labels the resulting integrals. Appendix A derives these index shifts. Analytic continuation gives the dimensionally regularized identity outside an initial convergence region [2].

A simple example shows how an identity becomes a reduction. A vacuum integral has no external momentum entering its denominators. Define its one-denominator example, called a tadpole, by

$$T_a(x) = \int \frac{d^d \ell}{i\pi^{d/2}} (\ell^2 - x + i0)^{-a}, \qquad x \neq 0. \tag{4}$$

Applying Eq. (3) with $v = \ell$ gives

$$0 = (d - 2a) T_a(x) - 2ax\, T_{a+1}(x), \qquad T_2(x) = \frac{d-2}{2x}\, T_1(x). \tag{5}$$

Appendix A shows the intermediate derivative. The last relation expresses the target T_2 through the master T_1. Its coefficient depends on d and the squared mass; reduction coefficients need not be constants.

2.3 What Kira calculates and where the work occurs

A seed is the index vector $\boldsymbol{\sigma}$ used in Eq. (3), and each seed supplies several derivative equations. Kira constructs a finite system, identifies relations using symmetry and linear dependence, and eliminates integrals according to an ordering. Back-substitution then writes the requested targets in terms of the remaining masters [4, 3]. The desired result is

$$I_{\boldsymbol{a}} = \sum_{m=1}^{n_M} r_{\boldsymbol{a}m}(\boldsymbol{x}, d) M_m, \qquad r_{\boldsymbol{a}m} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{6}$$

Here M_m is master m, n_M is the size of the chosen master set, and $\mathbb{Q}(\boldsymbol{x}, d)$ denotes rational functions with rational coefficients. Solving the linear identities gives this form, as shown in Appendix A. A smaller seed system can leave extra free integrals, so its remaining columns alone do not establish a complete master basis.

There are three distinct choices that a learned model can address: which seeds generate equations, which equations to retain, and which allowed row operation to perform next. A row is one equation in the coefficient matrix; an elimination step uses a nonzero coefficient to remove an integral from other rows. Appendix A derives that operation. Table 3 gives the models studied at these three locations and explains which remain active.

The coefficient arithmetic creates another cost. Symbolic elimination manipulates rational functions throughout the calculation. Alternatively, Kira can evaluate the system at numerical

points modulo a prime and use FireFly to reconstruct the rational functions [9, 10, 11]. Such a finite field uses exact arithmetic on residues, so the numerical samples are not floating-point approximations. Appendix R explains how samples constrain a coefficient. Our method can choose this existing arithmetic procedure, in addition to changing the equations or applying a direct identity.

3 Building the data and the first learned models

The project began with two questions: can a model find a sufficient set of seeds, and can it choose cheaper elimination steps? Both questions require reductions with known answers. We therefore constructed a collection of master bases, target expressions, source identities, and recorded reduction steps before testing the models inside Kira. Training adjusts a model's parameters using examples; testing evaluates the fitted model on reserved examples that did not supply those parameter updates.

3.1 From master integrals to training targets

The first collection contained 16 one- and two-loop families: tadpoles, bubbles, triangles, boxes, a two-loop propagator, a sunrise, and a vacuum sunset. Kira identified 49 master entries. We checked the selected bases by increasing the seed and probe bounds and changing the finite-field sample. All 48 identification calculations retained the same bases over those tested ranges. A master entry is counted within its family; a shared lower-sector integral can therefore appear in several entries. These checks concern basis stability within the tested ranges.

We then generated targets by starting with a dimensionally consistent combination of masters from one family,

$$\mathcal{T}_0 = \sum_{m \in J} c_m(\boldsymbol{x}, d) M_m, \qquad \mathcal{T}_{n+1} = \mathcal{T}_n + \lambda_n(\boldsymbol{x}, d) R_n, \qquad R_n = 0. \tag{7}$$

Here J is the selected set of master labels, c_m are rational coefficients, and R_n is a valid IBP, symmetry, or algebraic identity. The coefficient λ_n cancels a chosen term and replaces it with other integrals in the same family. Since the added expression is zero, the target keeps the known master coefficients. Appendix F derives this statement and the construction of individual integrals when an identity has only one unknown reduction.

Each record contains the family definition, target terms, master indices and coefficients, identities, seeds and derivative operators, and the allowed and selected elimination steps. Algebraic partial fractions carry no IBP seed and are identified separately. The recorded path provides a verified way back to the masters. It does not prove a globally shortest path: other identities or cancellations can give a different reduction.

The collection grew first to 1,000 family instances and then to 17,100 instances, with 57,979 master entries and 2,301,272 targets. The added high-loop examples primarily use the restricted coupled-vacuum family defined in Appendix G. Changing its masses gives a new family instance in this count, without producing a new graph structure. The collection spans loop orders one through twelve; later analytic and native test collections are separate.

The test part contains 148,494 individual integrals and 541,888 multi-integral expressions. The model receives the target and candidate choices, while the known answer, future reduction steps, and construction history remain outside its predictive inputs. This separation prevents the stored solution from answering the prediction task.

Table 1: The constructed learning collection. Related mass profiles remain in the same part of the grouped, approximately 70:30 training/test split across loop orders. Counts describe family instances, not distinct graph structures.

Quantity	Training	Testing
Family instances	11,963	5,137
Target expressions	1,610,890	690,382
Loop range	1–12	1–12

3.2 The seed and elimination models

The first seed model assigns a separate probability to every supplied candidate seed, so it can select several seeds for one target. It is a neural network, meaning a fitted sequence of numerical maps. It represents each target term, every ordered pair of terms, the family, the master indices, and the candidate seed. Shared 32-component maps and weighted term averages produce a candidate score. The set representation and weighted averages follow Refs. [12, 13]. The network has 28,097 fitted parameters and uses a threshold of one half to select seeds.

The elimination model instead receives a current coefficient state and its allowed operations. It scores each candidate from 132 features, combines shared 32-component representations with their mean and maximum, and assigns a probability over the allowed choices. It has 10,529 fitted parameters. Its original learning data describe one changing target row with fixed source identities; the subsequent test inside Kira supplies a different setting in which the matrix itself changes. We tested both settings separately.

Both models completed twenty further training passes over the expanded collection, continuing their saved parameters and AdamW optimizers [14]. A pass presents every eligible training example once. The seed model used all 1,610,890 training targets. The row model used 5,840,122 states with at least two choices; another 8,843,937 states have a forced step and supply no ranking decision. Binary and categorical likelihoods give the training losses derived in Appendix F. The notes retain the earlier training phases and detailed architectures.

3.3 Why prediction accuracy was only the first test

Seed-label agreement measures whether a model reproduces the recorded reference set. Algebraic sufficiency measures whether its selected equations actually reduce the target to the prescribed masters. Elapsed time measures whether the saved work exceeds prediction and recovery costs. We evaluated these three questions in that order.

The held-out seed test gave 95.750% precision and 96.892% recall, with complete-set agreement of 84.332%. Precision is the fraction of selected seeds that carry reference label one; recall is the fraction of reference seeds that were selected. Their harmonic mean, F_1, is 96.318%. Appendix F defines these quantities. The selected equations were sufficient for 652,517 of 690,382 targets, or 94.515%; the complete candidate pool was sufficient for all of them. Figure 1 separates these results.

The original row model completed 99.988% of constructed test targets, compared with 99.901% for the deterministic greedy rule. Its average decision-path time was nevertheless 12.344 milliseconds per target against 0.699 milliseconds, about 17.7 times as long. High reference agreement therefore did not establish acceleration. This result motivated the later changes in model objective and integration.

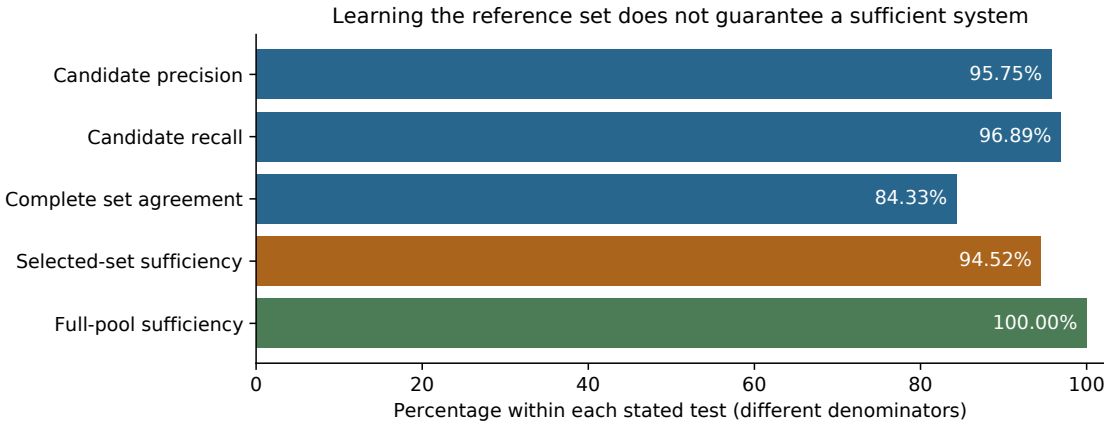


Figure 1: The initial seed model learns useful reference sets, but exact sufficiency requires a separate test. All percentages use the frozen model and the 690,382 reserved targets. Label precision and recall count candidates; set agreement and sufficiency count targets. The complete supplied pool resolves every target. These are different measures, not successive stages of one probability.

4 From learned choices to a combined reduction method

The initial models showed that reproducing a reduction path and reducing its cost are separate tasks. We therefore changed both the learning objective and the point where a decision enters Kira. The resulting procedure first tests the mathematical conditions for an exact identity. It applies an accepted formula to a whole target or to a reducible part, then uses Kira for the remaining integrals. Within that calculation, a learned choice is used only in the domain where its validation supports it. This ordering gives each method a defined role and allows a direct comparison with the same calculation performed without the additional learned choice.

4.1 Moving the decisions to the work they can avoid

A seed model can save equation generation only when it acts before that generation. We therefore progressed from reductions with stored identities to filters inside Kira's native generation, and placed the equation classifier before full coefficient construction. We also tested the row model inside the changing elimination matrix, then reduced its inference overhead with a compiled cost policy. The tests in Section 6 determine which of these interventions were useful.

The next stage predicted complete measured cost for a target batch. Candidate choices included smaller or larger seed regions, limits on numerator and extra denominator powers, equation filters, integral orderings, and the number of arithmetic threads. We compared linear regression, trees, forests, and neural regressors, choosing their settings on separate validation inputs. Some models specialized to a graph structure; a later shared model used propagator and index features without family-name inputs. Appendix J defines that model, its ten actions, and its objective.

We also trained a strategy selector, which we call a portfolio model. It selects a complete procedure, including an earlier policy or selective numerator expansion and a worker count. Its twenty features describe graph structure, target powers, numerator ranks, target count, and the distribution of extra denominator powers. Thirty candidate trees and ensembles used measured relative times as labels, and validation selected both a per-graph fixed control and any enabled predictor. The native algebra and recovery rules stayed common to the compared methods. Later preparation changes reused valid symmetry and equation work while retaining the same selected strategies.

The prescribed master basis also affects cost. We added seed bounds that contain its indices, selective increases when a target needs further identities, and a two-stage calculation through

an intermediate basis. If the target coefficients in that basis are C_{ij} and its conversion to the required masters is B_{jm}, substitution gives the final coefficients $\sum_j C_{ij}B_{jm}$. Appendix F derives this composition. Every second-stage calculation and recovery attempt enters the complete cost.

Finally, exact structural tests precede learned selection. An accepted formula can remove the entire native calculation, and independent loop blocks can share component reductions. A model is then evaluated only for the remaining choices where validation supports it. This organization avoids prediction on a request already resolved by an exact identity.

4.2 Exact formulas and recurrences

The tadpole relation in Eq. (5) illustrates the first route: a recurrence repeatedly lowers an integer power. A direct formula can go further and provide an analytic value. In our normalization, Appendix O derives

$$T_a(x) = (-1)^a (x - i0)^{d/2-a} \frac{\Gamma(a - d/2)}{\Gamma(a)}. \tag{8}$$

The Gamma function is defined for $\operatorname{Re} z > 0$ by $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt$ and elsewhere by analytic continuation [15]. It satisfies $\Gamma(z + 1) = z\Gamma(z)$, so integer shifts of its argument give finite products. These products explain why the ratios of the target and master values in our analytic families are rational.

Some formulas depend on a particular physical limit. Feynman parametrization combines denominator factors through an auxiliary parameter integral. For the bubble of Eq. (2) at $p^2 = 0$, this procedure gives

$$B_{11}(0; x, y) = \frac{T_1(x) - T_1(y)}{x - y}, \qquad B_{ab}(0; x, y) = \frac{\partial_x^{a-1} \partial_y^{b-1} B_{11}(0; x, y)}{(a - 1)!(b - 1)!}. \tag{9}$$

The first expression assumes $x \neq y$. At equal nonzero masses the limiting result is $B_{ab}(0; x, x) = T_{a+b}(x)$. Appendix M derives both cases and the extension to polynomial numerators. This separate formula avoids dividing by a matrix of external scalar products that becomes singular at this kinematic point.

A further exact rule treats a massless triangle with three denominator powers a, b, c, two lightlike external momenta p_1, p_2, and one nonzero invariant $Q = -(p_1 + p_2)^2 > 0$. Lightlike means $p_1^2 = p_2^2 = 0$. Appendix S defines the integral $T_{abc}(Q, d)$, derives its Gamma-function representation, and converts it to a bubble master. In particular,

$$T_{111}(Q, d) = \frac{2(d - 3)}{Q(d - 4)}\, B(Q, d), \qquad B(Q, d) = T_{101}(Q, d). \tag{10}$$

This is a standard analytically tractable kinematic configuration [16]. We derive the coefficient with our sign and normalization conventions before using it. We first used an analytic triangle master and later expressed it in a bubble basis with an explicit master value.

An early exact result came directly from the high-loop training family. Its massive denominators all contain the same loop-momentum sum. Euler identities and partial fractions give a power-lowering recurrence whose loop dependence enters through $z = L(d - 2)$. We compute its coefficient polynomials once and substitute each requested loop order. Appendix G proves the recurrence and an independent check. Section 6.3 reports its original-to-optimized cumulative comparison.

The same reasoning extends to selected many-loop structures. If separate groups of loop momenta occur in separate factors of the integrand, their reductions multiply. Some connected families also allow successive bubble integrations from the outermost loop inward. Appendix N defines these trees and derives their Gamma products. Appendix A proves the independent-product rule. The accepted domains are summarized in Table 2; their restrictions prevent the formulas from being applied to a different physical problem.

For clarity, a scalar denominator integral has no free Lorentz index. Polynomial numerators arise after contractions of tensor integrals, whose numerators carry momentum components. We use angular averages to reduce these numerator terms; Appendix L derives the required contractions. An integral is scaleless when it has no nonzero mass or external invariant that can supply a dimensionful scale. Appendix O derives its zero value in dimensional regularization. The imple-

Table 2: Exact methods used in the application. These are implementation domains, with derivations in the stated appendices. "Rank" is the total numerator power, and "excess power" counts positive powers beyond one.

Integral structure	Method	Principal conditions
One-loop scalar and polynomial-numerator sectors	Angular reduction and a power-lowering recurrence (Appendix L)	At most five coordinates; rank and excess power at most six; required matrices invertible
Two-point sectors at $p^2 = 0$	Endpoint tadpoles and mass derivatives (Appendix M)	Rational tensor inputs: rank and excess power at most four; scalar symbolic masses: individual powers at most six and total at most eight
Vacuum factors with one momentum per factor	Partial fractions and tadpole values (Appendix O)	Nonnegative rational squared masses; at most eight denominators per factor; powers bounded as stated there
Scaleless vacuum families	Zero in dimensional regularization (Appendix O)	No mass or external momentum scale; homogeneous quadratic denominators
Connected bubble trees	Successive exact bubble integrations (Appendix N)	Massless two-point root or one massive vacuum root; no additional occupied auxiliary lines
Massless triangles	Direct coefficient and Gamma value (Appendix S)	Two lightlike legs; rational $Q > 0$; powers 0–4 and total at most eight
Independent products	Product of component reductions (Appendix A)	Every occupied factor belongs to an accepted independent component

mentation admits the listed vacuum, tree, and independent-product structures through thirteen loops. This loop range describes specific structures. Additional propagators that couple their loop momenta can invalidate the factorization or the tree integration. Unsupported inputs therefore continue to the remaining exact rules or Kira, subject to the application's resource bounds.

4.3 Learned choices within Kira

Machine learning enters as a prediction of a useful computational choice. Training fits a model to examples, validation selects its settings on separate examples, and testing evaluates the frozen choice on reserved inputs. A model receives quantities calculated from the target indices and propagators, called features, and estimates the cost or usefulness of an allowed action. The algebra still determines the reduction coefficients. A decision tree splits its inputs through a sequence of feature tests and returns a prediction at the final branch [17]. Table 3 separates the original three interventions. Here and below, *native* means that Kira generates and solves the IBP system.

The retained seed model selects among wide, medium, and tight regions of seed indices. It was trained on 34 batches and selected using sixteen separate validation batches. Appendix B defines

Table 3: The three learned interventions and their current use. The later strategy models and their measured effects are compared in Table 5. The notes retain the earlier architectures and training histories.

Choice	Model and input	Predicted action	Final application
Seed selection	A depth-two cost tree, with twelve depth-three failure trees per action; family and target features	One of three seed regions	Retained in its accepted domain
Equation selection	A decision-tree classifier; seed, target, and derivative features	A retention score for each seed–operator pair	Disabled in the final comparison
Row selection	A set-based neural ranker for the current matrix and allowed steps; later a compiled cost policy	The next elimination step	Omitted; native row rules retained

the regions. They are candidate searches, so a predicted choice does not prove that its equations suffice.

The equation model has seventeen features that describe a candidate seed, its derivative operator, and its relation to the targets. Its training labels identify equations retained by a complete reference reduction. A score threshold selects equations before their full coefficients are constructed. Appendix B specifies the features and weighted training labels. The original row model uses a neural network: a fitted sequence of numerical maps that assigns each candidate step a score. It combines 132 candidate features through shared 32-component representations, their mean and maximum, and a map with 64 intermediate components. Combining a set of candidates in this way follows the set-based representation of Ref. [12]. The notes give its training on constructed target-row reductions and its subsequent test inside Kira's changing matrix. A row model acts when the matrix already exists, and must save enough algebra to cover the cost of choosing each step. The measured row results did not support its use in the final application.

4.4 Learning the cost of the coefficient arithmetic

We also tested a model that chooses the arithmetic for the remaining native reductions. The candidate actions are the preceding procedure and finite-field reconstruction with 1, 4, 8, or 16 evaluation points processed together. This group size is separate from the number of target integrals in a request.

The final arithmetic-selection training collection contains 48 requests across four propagator structures, with 20 further requests for validation. A three-loop training comparison did not agree with its reference; later checks did not reproduce the discrepancy. We therefore keep the preceding procedure for that structure and fit the added model to the 36 two-loop training requests. The three-loop requests remain in the complete comparison. The notes preserve the initial failure and the subsequent checks.

The fitted model averages 32 extremely randomized regression trees, using the construction of Geurts, Ernst, and Wehenkel [18]. Its 29 features describe the loop count, masses, denominator connections, target powers, and sizes of the supplied rational kinematic numbers. It predicts complete time relative to the same request's preceding procedure. Appendix C derives the weighted training objective and lists the features.

Validation selects tree depth from 2, 4, and 6, minimum terminal-branch population from 2 and 4, and the required predicted saving from zero, five, and ten percent. A fixed control chooses its best setting separately for each of the four structures. The model is enabled only if it reduces validation cost by at least three percent and selects no failed procedure. This is an experimental

acceptance condition, not a mathematical guarantee for unseen integrals. The model fails this condition in the recorded comparison, so the final application uses the fixed settings and keeps the earlier seed model.

4.5 Completion, recovery, and shared work

An accepted exact formula requires its stated propagator and kinematic conditions. A restricted native calculation must express every target in the required master basis. When it does not, the application attempts its configured broader or preceding procedure and includes the spent time in the result. In the experiments, a separate exact comparison also subtracts every returned coefficient from the reference coefficient in the same basis. This additional check tests the actual output; a model score alone supplies no correctness certificate.

We also tested joint reduction of compatible target lists, exact reuse across different physical scales when justified, and processing a list of requests in one call. This shares initialization and repeated exact component calculations across the list. It preserves the factorized form of independent products instead of expanding all products of master terms. The final timings include this shared-call organization in both the new method and its control, as explained next.

5 Comparison methods

The study contains several workloads because the questions changed as the method developed. Constructed targets test whether learned decisions can recover known reductions. Native tests ask whether they save work when Kira must generate the equations. Exact-family tests ask how much algebra a proved identity can replace. We retain these distinctions when reporting the gains.

5.1 Controls and timing boundaries

A native control uses Kira's own choices with the same physical inputs and the stated seed bounds. A fixed control uses a strategy selected from development data, without request-dependent prediction. These controls share the surrounding implementation with the learned method. A comparison with an earlier version of our application instead measures the combined effect of changes to formulas, preparation, models, and execution. The restricted-vacuum cumulative test has its own original supplied-equation Kira control. It does not measure native generation of a general multiloop IBP system.

The reported time includes the operations inside each experiment's declared boundary, with all unsuccessful attempts and recovery included. The earliest supplied-equation test excludes construction of its stored identity pool. Native model tests include equation generation and exact elimination. Later complete-application tests also include process startup, loading, prediction, output, and cleanup. External reference preparation usually lies outside production time; the early native tests include coefficient comparisons, while the later application tests perform that audit outside the timer. The cumulative vacuum test includes output reading. These boundary differences remain part of the result, rather than being hidden in one global percentage.

Most native experiments use a prescribed master basis. Their times therefore concern subsequent reductions to that basis. We separately investigated basis discovery and compact preparation, retaining their costs. Offline training and the reductions used to produce training labels remain outside application timing. For example, the shared structural model required 9.5 seconds to fit after existing labels were collected; preparing and checking its 110-request reference grid required 1,056.4 and 772.2 seconds, including unsuccessful attempts. These costs matter when deciding how often a model or prepared basis must be reused.

5.2 Independent examples and exact checks

The constructed-data split groups related mass profiles across loop orders. Later native fits separate training, validation, and reserved parameter profiles or target batches. Some tests change parameters within known graph structures, while others exclude complete physical groups or loop orders from fitting and selection. A new parameter point does not constitute a new graph, and an independent product of many loops does not test a general connected graph at that loop order.

The reducer receives the family, targets, and required master indices, without the target's reference coefficients. Exact coefficient subtraction or polynomial cross multiplication checks the returned answer against an independently prepared reference in the same basis. Where bases differ, an exact conversion is required before equality can be asserted. Algebraic completion requires every requested target to reduce to the prescribed basis. Analytic paths additionally require their propagator and kinematic conditions, which are proved in the appendices and checked before substitution.

All failures remain in the coverage counts. Timing ratios over matched completions exclude unverified pairs and state how many pairs remain. In particular, the 43-request shared-model timing result comes from a 110-request grid, while the final 93-request serial comparison completes every requested output. A coordinate limit or a bounded reference failure does not prove that the corresponding integral is mathematically irreducible.

5.3 Estimators and the meaning of a cumulative gain

For paired methods, a time ratio below one means less elapsed time. We use either a ratio of summed request medians, an equal-group geometric mean of paired ratios, or the median time of one complete collection. Appendix E defines the current estimators; Appendix H derives the geometric and total-time forms used in the earlier comparisons. Every table identifies its estimator. Intervals retain the original experiment's paired bootstrap units, following Ref. [19]; repeated runs measure execution variation and do not create additional independent families.

A cumulative gain requires the original and final methods to be measured on identical requests with the same timing boundary. The eight-group vacuum comparison and the 75-request direct-vacuum-plus-reconstruction comparison satisfy this requirement for their respective stages. The subsequent 93-request comparison has a different population and preceding control. Multiplying their speed factors would describe no experiment that we performed.

Appendix U gives the full final protocol: Kira 3.1 with FireFly, one computational thread, three rotated repetitions on the same node, 20 connected native requests at two and three loops, and 73 analytic requests from restricted structures through thirteen loops. The native input definitions and coordinate limits appear in Appendix K.

6 Results

We obtained three kinds of result: learned reductions in native Kira cost, exact shortcuts for specified integral structures, and savings from sharing work between compatible requests. The following comparisons identify the control and workload for each result. Their ratios describe separate experiments and are not factors in a single universal speedup.

6.1 Seed, equation, and row selection inside Kira

Using the first seed model with already supplied identities reduced time by 8.6% on twelve single-target batches: the model-to-full-pool ratio was 0.914, with interval $[0.871, 0.958]$. Generating

that identity pool was outside this timer. The corresponding sixteen-target comparison had ratio 0.952 with an interval crossing one, so the larger-batch gain was unresolved.

We then placed the learned decisions inside native equation generation and elimination. This test included the work absent from the supplied-equation comparison. The original row model increased complete time by 30.5% in 84 trials. A later compiled row-cost policy gave ratio 0.982 with interval $[0.959, 1.003]$ on sixteen batches from eight physical profiles. Its interval includes equal time. We therefore omitted learned row selection from the retained application.

Equation selection gave a clearer native result. Filtering seed–operator pairs before constructing their coefficients reduced equation-preparation time by 36.2% and complete time by 5.7% in the original controlled test. The complete-time ratio was 0.943 with interval $[0.915, 0.970]$. A later test on 24 new-index batches gave ratio 0.933 with interval $[0.928, 0.940]$, or 6.7% less complete time. The later retained seed policy gave ratio 0.963 with interval $[0.937, 0.984]$ on 32 box batches, or 3.7% less time. These comparisons support learned selection on their stated inputs, while their different controls prevent addition of the percentages.

Transfer to other graph structures exposed a limit. On nonplanar boxes, pentagons, and boxes with external virtuality, the original equation filter needed recovery and took 25.2% more time in the first transfer study. Training separate seed and equation trees on those structures removed every first-attempt failure in the new reserved comparison. Their time ratios to native choices were still 1.027 and 1.017, with intervals including one. Learning sufficiency solved one difficulty without establishing a total-time gain.

6.2 Training for complete cost and the required master basis

We next trained models on measured reduction cost. The initial common strategy predictor changed time little relative to an equally prepared native control. Separate strategies for seven known graph structures performed better: their aggregate ratio to native choices was 0.937, and the ratio to the per-graph fixed control was 0.944. This second comparison isolates the added request-dependent selection more closely.

Selective seed expansion and reduction through an intermediate master basis then gave a ratio of 0.852 to native choices and 0.926 to the matched fixed strategy on 36 batches. With equal weight for each graph, this corresponds to 14.8% and 7.4% less time, respectively. Summed batch medians were 106.24 seconds for native choices and 77.03 seconds for the selected method. The 27.5% reduction in that sum differs from the 14.8% equal-graph result because long requests receive more weight in the total. All 720 timed reductions passed the exact comparisons.

We also tested alternative integral orderings, the number of arithmetic threads, wider action portfolios, and models that used structural features rather than family names. The ordering extension did not improve its preceding method. Thread selection improved the preceding method by 1.8%, but its interval against the matched fixed control included equal time. These tests show why each additional learned choice requires its own comparison.

A portfolio model chooses a complete reduction strategy from a defined set. Our five choices combined earlier policies or selective numerator-rank expansion with one or two Kira workers. Trees selected from 47 training and 26 validation batches were tested on 26 further batches from nine known structures through four loops, including symbolic scattering inputs. Total time was 85.497 seconds against 108.029 seconds for native choices and 86.809 seconds for the per-graph fixed portfolio. The gain against the stronger fixed control was therefore 1.5%, compared with 20.9% against native choices.

We then retained those model decisions and reduced repeated symmetry preparation and equation construction around them. On another 26-batch collection, selected-method time fell from 79.784 to 64.183 seconds, a 19.6% saving from the integration changes. Native choices with the same improved preparation took 87.847 seconds, and the matched fixed portfolio took 67.381

seconds. These controls give savings of 26.9% and 4.7%, respectively. All 312 timed reductions passed the exact checks. A fixed portfolio can still contain an earlier learned policy, so this comparison isolates the added outer selection rather than removing every model.

The shared structural model supplies a separate generalization result. It used 44 features and a neural network with two 48-component hidden layers to select among ten seed configurations. On 43 matched native requests, complete time fell from 63.450 seconds for native choices to 57.125 seconds, or 10.0% less time. The fixed control took 60.094 seconds, so the added learned choice saved 4.9%. Fifteen matched requests came from excluded physical groups or loop orders; their learned-to-fixed ratio was 0.885. The model and physical inputs are specified in Appendix J.

Table 4: Selected native comparisons in this study. Each ratio divides the selected method's time by its stated control. Intervals retain the original experiment's paired resampling method; graph-weighted and summed-time estimators are distinct.

Intervention	Control and estimator	Ratio	Interval
Equation filter, original test	Native; equal physical groups	0.943	[0.915, 0.970]
Equation filter, new indices	Native; fixed-request repetitions	0.933	[0.928, 0.940]
Graph-specific strategy	Per-graph fixed; equal graphs	0.944	[0.899, 0.986]
Selective expansion and basis conversion	Fixed strategy; equal graphs	0.926	[0.876, 0.975]
Shared structural model	Global fixed; summed medians	0.951	[0.916, 0.971]
Retained box seed policy	Fixed seed region; summed medians	0.963	[0.937, 0.984]
Compiled row policy	Native rows; grouped geometric mean	0.982	[0.959, 1.003]
Learned choices after reconstruction	Fixed choices; summed medians	0.983	[0.951, 1.003]

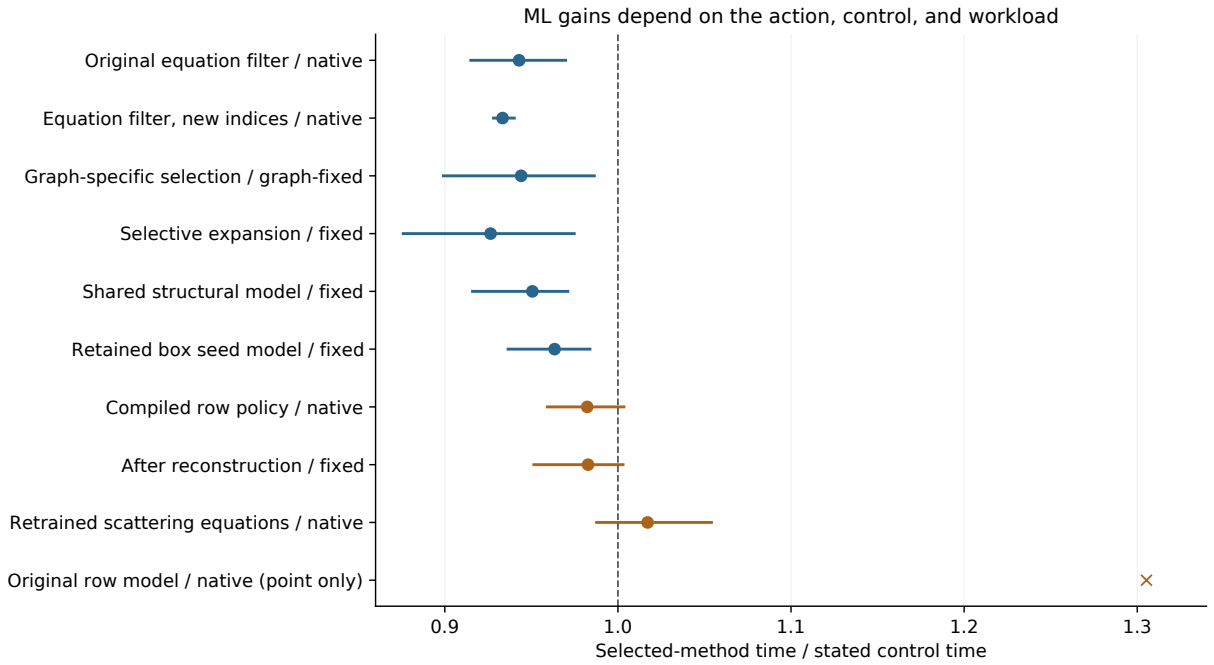


Figure 2: We measured gains from learned choices on some workloads and retained unresolved or slower results. Each point and interval comes from the named experiment and control in Table 4. The original row and adapted scattering results are included to show the rejected approaches. The equal-time line is one. These separate comparisons do not form a cumulative sequence.

The shared-model grid contained 110 requests, of which only 43 produced a complete matched native comparison. Twenty-one initial references and ten enlarged-region checks failed, and 36 inputs exceeded the tested native coordinate limit. Completed comparisons reached five loops, with only two matched requests at five loops; their aggregate learned-to-native ratio was 1.172.

Figure 4 keeps the full requested range visible. The 4.9% learned gain therefore applies to the matched set and does not establish successful acceleration throughout the grid.

Table 5: Models we developed and the measured reason to use or omit them. Each row names its own control. The final application retains a validated seed policy, while the other rows report distinct experiments in the study.

Model and representation	Decision and advantage	Measured result and limitation
Seed-set neural classifier; 28,097 parameters	Independent seed probabilities allow several seeds per target; learns from exact construction records	94.515% sufficient predicted sets on 690,382 constructed test targets. This accuracy does not establish a native runtime gain
Row neural ranker; 10,529 parameters, 132 input features	Scores allowed elimination steps from the current state	30.5% slower in 84 native trials. The later compiled cost policy gives ratio 0.982 with interval $[0.959, 1.003]$. Omitted
Equation decision tree; 17 seed, operator, and target features	Rejects unlikely useful equations before full construction	72.295 → 67.478 s on 24 new-index batches: 6.7% less time. Later nonplanar filtering causes recovery on all attempts; disabled in the final test
Graph-specific cost trees and strategy selectors	Choose seed enlargement, basis conversion, or a complete strategy from measured time	Selective expansion and basis conversion save 7.4% against a per-graph fixed choice, with equal graph weighting. Requires training and validation for the tested structures
Shared structural neural regressor; 44 inputs, two 48-component layers	Predicts ten seed-action costs without a family-name input	60.094 → 57.125 s against a global fixed choice: 4.9% less time on 43 matched requests. Some classes are slower; Table 9 gives every class
Native seed-cost trees with failure checks	Choose a seed region, or use a conservative fixed action when risk is high	Earlier 32-box test: 156.383 → 150.663 s, 3.7% less than fixed seeds. Later 24-box test: 200.983 → 198.313 s, 1.3% less, with interval including equal time
Tensor-order cost tree and later arithmetic forests	Attempt to choose an exact recurrence order or native arithmetic setting	Validation did not justify deployment; final methods use fixed choices. Exact-method gains are reported separately

6.3 The large cumulative gain from exact vacuum recurrences

The constructed high-loop data also led to a mathematical shortcut. In the restricted coupled-vacuum family, every massive denominator depends on the same momentum sum, and all massless lines have power one. Euler identities and mass partial fractions reduce every accepted target to one-power masters. Their coefficients depend on the loop order through the single variable $z = L(d-2)$, so one exact polynomial calculation serves several loop orders. Appendix G derives the recurrence and an independent Taylor partial-fraction check.

We tested the cumulative change directly against the original saved Kira 3.1 executable on eight new groups, each with three mass profiles and all loop orders one through twelve. Both baseline arms constructed the complete supplied Euler and mass-identity systems for each mass profile and loop order. The exact method used the fixed recurrence, shared loop dependence, and exact coefficient expansion. A learned recurrence selector was trained and tested, but the final validation-selected rule was fixed. Thus this large gain is an exact-method and work-sharing result.

Summing one complete-time median per group gives 1,027.781 seconds for the original baseline and 1.922 seconds for the exact method, a 534.9-fold speedup. The equally weighted geometric mean of the eight speedups is 543.8, with descriptive interval $[488.1, 605.1]$. Both measures include fresh-process startup, input preparation, equation construction where required, exact algebra, checking, and output writing and reading. All 120 workflows complete; each baseline arm makes 1,440 Kira calls over its repetitions, while the successful exact arm makes none. Independent checks cover 27,960 nonzero coefficient polynomials.

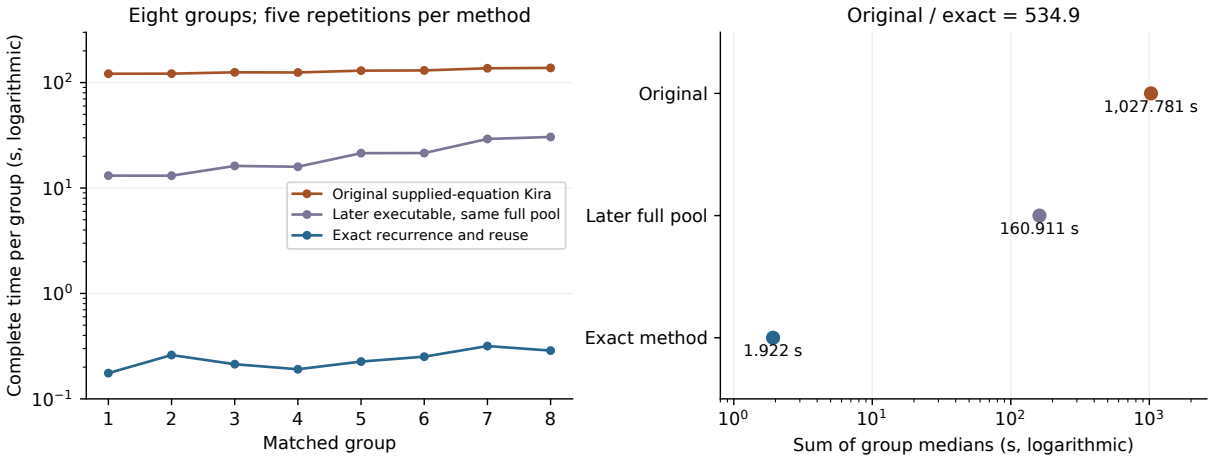


Figure 3: A direct original-to-optimized comparison for the restricted vacuum family. Left: median complete time for each of eight matched groups on a logarithmic axis. Each group contains three mass profiles and loop orders one through twelve. Right: sums of group medians, also on a logarithmic axis. The original and later executable controls solve supplied identities separately; the exact recurrence shares their symbolic solution. The 534.9-fold total-time ratio is not a model-only gain or a result for general scattering.

6.4 Sharing requests and extending the analytic domains

Another saving comes from combining target lists that have identical family definitions and kinematics. On ten compatible request groups, joint reduction with native choices and the common implementation had ratio 0.517 to separate native reductions, with interval $[0.507, 0.535]$: 48.3% less time. Adding the learned choice gave ratio 0.996 against that joint fixed control, with an interval crossing one. Linearity permits the shared answer to be separated back into the original requests, as derived in Appendix F.

We also separated independent loop blocks, reused exact component reductions, and replaced suitable native calculations by analytic identities. The resulting domains include massless and one-mass two-point structures, connected bubble trees, one-loop tensor numerators, zero-invariant bubbles, vacuum partial fractions, and massless triangles. We derived their conditions before applying them, and tested restricted connected trees and independent products through thirteen loops. These additions are summarized in Table 2; their derivations remain in the appendices.

The development included compact basis preparation, failure-aware choices, basis-compatible seed bounds, symbolic factorization, cancellation of Gamma factors, and delayed model loading. Each change addressed a measured cost. For example, compact preparation without ML had ratio 0.305 to expanded discovery on thirteen matched requests, while the added neural choice had ratio 0.982 to the compact fixed procedure on 22 requests and did not resolve an additional gain. In another complete-cost experiment, the method completed 126 of 127 requests and saved 6.0% against the preceding application, but gave no resolved extra gain against the matched fixed control. These outcomes prevented setup savings and successful recovery from being reported as model-only acceleration.

6.5 Later combined results and the final application

Table 6 collects complete-time results whose baselines were measured directly. They answer different practical questions: repeated reductions of an analytically reducible family, compatible requests that can share a system, a native model choosing a seed configuration, and mixed analytic/native workloads.

Table 6: Measured gains with explicit controls. Each row is a separate matched experiment. Times are sums of request or group medians, except the final row, which uses the median time of a complete collection. No row combines percentages from different tests.

Workload and control	Control (s)	Method (s)	Measured gain
Restricted vacuum; original supplied-equation Kira	1,027.781	1.922	534.9-fold; exact recurrence and reuse
36 native batches; native choices	106.24	77.03	27.5% less total time; selective expansion and basis conversion
26 native batches; equally prepared native choices	87.847	64.183	26.9% less; integrated strategy portfolio
Same 26 batches; fixed portfolio	67.381	64.183	4.7% less; added outer selection
43 native requests; global fixed choice	60.094	57.125	4.9% less; shared learned choice
75 mixed requests; before direct vacuum and reconstruction additions	81.827	49.852	39.1% less; combined additions
93 mixed requests; preceding integrated application	45.091	41.137	8.8% less; final update

On the 75-request collection, enabling the validated finite-field reconstruction settings reduced time from 79.624 to 49.852 seconds, or 37.4%. Comparing the same final result directly with the application before both the direct-vacuum and reconstruction additions gives 81.827 to 49.852 seconds, or 39.1%. Those are matched comparisons on the same collection; they are the relevant cumulative measurements for that stage.

The later 93-request study gives the additional 8.8% reduction from 45.091 to 41.137 seconds. It contains 20 native and 73 analytic requests and checks all 837 outputs from three methods and three repetitions. This is the last incremental comparison, rather than the overall outcome of the study. Appendix V retains its full timing tables, three plots, subgroup regressions, and failed parallel predecessor.

The final application combines admitted exact identities, shared work, the inherited validated seed model, and fixed native arithmetic choices. The original row model remains omitted. The latest arithmetic selector failed its validation condition, and the added nonplanar equation filter was 71.1% slower after recovery; both remain disabled. In the reconstruction-era model-only comparison, the ratio to fixed choices was 0.983 with interval $[0.951, 1.003]$, so a reliable extra aggregate ML gain was unresolved. Earlier positive model results remain valid for their own tests; they do not make every later configuration faster.

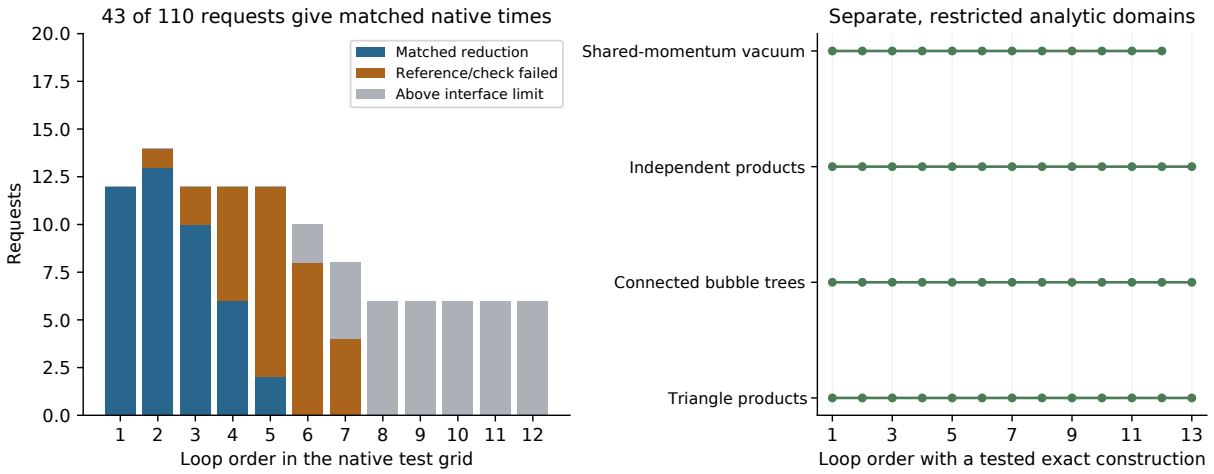


Figure 4: Loop count and mathematical structure define different limits. Left: all 110 requests in the shared-model native test, including failed reference checks and inputs above the tested interface's coordinate limit. Right: the loop ranges demonstrated by selected exact constructions in separate studies. The higher-loop analytic results concern the stated restricted structures, not all connected graphs. No native timing at loop orders six through twelve is inferred from an analytic result.

7 Results by integral structure and physical use

The preceding comparisons measure different parts of the reduction. We now connect them to the integrals that each method accepts. Table 7 gives the final analytic expressions, the corresponding physical context, and a directly measured gain. Tables 8 and 9 separate the one-loop tensor and shared-model results by integral type. These tables describe reductions of specified integral families; we did not calculate a complete scattering amplitude or decay rate in this study.

The physical examples explain why these mathematical structures matter. A two-point integral depends on one external momentum and enters propagator corrections; three-point and four-point integrals enter vertex and scattering corrections, respectively [1, 7]. Massive vacuum integrals also enter heavy-quark threshold matching in QCD after external momenta are set to zero [20]. Five-point integrals occur in amplitudes for five external partons [21]. Our particular mass choices and reducible tree structures are test inputs within these broader classes. The application does not infer that a formula covers a full physical process from the number of loops or external particles alone.

We report seconds and a speed factor $S = T_{\mathrm{control}}/T_{\mathrm{method}}$. Thus $S > 1$ means a faster calculation, and the percentage time saving is $100(1 - 1/S)$. These are the reciprocal and percentage forms of the time ratio defined in Appendix E. Each factor below comes from one matched experiment. Except for the explicitly identified original-vacuum comparison, the analytic controls are preceding versions of our application and already contain earlier improvements. Offline training and external reference preparation remain outside these production times.

Table 7: Analytic methods, physical uses, and measured gains. Each formula uses the integral normalization defined in its cited equation or appendix. $d = 4 - 2\epsilon$ throughout. Times sum matched request medians, or group medians for the original vacuum test. The gains in this table come from exact algebra and shared work; none is an isolated ML gain.

Integral and physical context	Final formula and reduction method	Measured control and gain	Applicability and limitation
Massive tadpoles and independent vacuum factors. Vacuum integrals occur in heavy-quark decoupling calculations [20].	$T_a(x) = (-1)^a (x - i0)^{d/2-a} \dfrac{\Gamma(a - d/2)}{\Gamma(a)}$. x is a squared mass. Partial fractions reduce several masses carrying the same loop momentum to these tadpoles. Independent factors multiply. We derive the normalization and repeated-mass limits in Appendix O.	Thirteen two-mass product requests, loops 1–13: preceding application 2.271 s; direct method 1.935 s. Speedup 1.17×; 14.8% less time.	Provides analytic values as well as master coefficients. The measured high-loop targets are independent products. A general coupled vacuum graph requires additional identities. No separate standalone-tadpole speed factor was measured.
One-loop scalar and polynomial-numerator bubbles, triangles, and boxes. These enter two-, three-, and four-point corrections [7].	$I_{\boldsymbol{a}} = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) I_{\boldsymbol{a}_S - \boldsymbol{t}}$. S contains the positive denominator indices. Angular averaging gives the finite numerator polynomial $\sum_{\boldsymbol{t}} c_{\boldsymbol{t}} \prod_{i \in S} D_i^{t_i}$; each term cancels denominator powers. Equation (63) gives every angular coefficient, and Eq. (71) lowers the remaining raised powers.	Twenty one-loop requests: 53.518 → 6.343 s against the preceding application. Speedup 8.44×; 88.1% less time. Twelve independent tensor products, loops 2–13: 121.891 → 3.006 s, or 40.5×.	We derived the angular reduction and used factored dimension denominators. Required Gram or Cayley matrices must be invertible; the tested inputs have rational kinematics and symbolic d. Validation rejected the tensor-order model, so these gains use a fixed exact path.
Two-point sectors at $p^2 = 0$. These are zero-invariant limits of self-energy integrals and can occur as lower sectors of larger families [1].	$B_{11}(0; x, y) = \dfrac{T_1(x) - T_1(y)}{x - y}$, $B_{ab} = \dfrac{\partial_x^{a-1} \partial_y^{b-1} B_{11}}{(a-1)!(b-1)!}$; $B_{ab}(0; x, x) = T_{a+b}(x)$. Here x, y are squared masses. Appendix M derives the scalar and numerator formulas.	Twenty tensor requests: 27.055 → 5.856 s, or 4.62× and 78.4% less time. Three symbolic-mass scalar requests: 4.564 → 3.958 s, or 1.15×. Controls use the preceding application.	Avoids a singular external Gram-matrix inversion at this limit. The equal-mass expression supplies the finite limit of the difference quotient. The rule requires the stated invariant and power conditions; it does not apply at generic p^2.

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Table 7 continued

Integral and physical context	Final formula and reduction method	Measured control and gain	Applicability and limitation
One-scale massless triangles and their independent products. The two-lightlike-leg configuration occurs in on-shell vertex calculations [16].	$T_{abc} = (-1)^A Q^{h-A} \frac{\Gamma(A-h)\Gamma(h-b-c)\Gamma(h-a-b)}{\Gamma(a)\Gamma(c)\Gamma(d-A)}$. $A = a+b+c$, $h = d/2$, $Q = -(p_1+p_2)^2 > 0$, and $p_1^2 = p_2^2 = 0$. In a bubble basis, $T_{111} = \frac{2(d-3)}{Q(d-4)} B$, with $B = T_{101}$. Equation (116) gives the general integer-power coefficient. For independent components, $\prod_j T_j = (\prod_j R_j) \prod_j B_j$.	Thirteen independent triangle-product requests, loops 1–13: 55.253 → 2.540 s, or 21.75× and 95.4% less time, against the preceding application. With the earlier reference preparation already supplied, its control takes 29.240 s, giving 11.51×.	Provides explicit values or rational coefficients in a bubble basis. The measured comparison belongs to the earlier analytic-batch stage; the final triangle-to-bubble derivation fixes the later basis. These timings do not isolate that basis change. No coupled multiscale high-loop triangle graph is covered.
Restricted coupled-vacuum family with shared massive momentum. This is our special benchmark within the broader class of QCD vacuum integrals [20].	$V_{\boldsymbol{b}}^{(L)} = \mu^{1-B} \sum_j C_{\boldsymbol{b}j}(L(d-2)) M_j^{(L)}$, $C_{\boldsymbol{b}j}(z) = \sum_{k=1}^{b_j} a_{jk} \prod_{h=1}^{k-1} \frac{2h-z}{2h\rho_j}$, $a_{jk} = [t^{b_j-k}] \prod_{i\neq j} (\rho_i - \rho_j + t)^{-b_i}$. $\mu_j = \rho_j \mu$, $B = \sum_j b_j$, and $M_j^{(L)} = V_{\boldsymbol{e}_j}^{(L)}$. The brackets select a Taylor coefficient. Appendix G defines the family and derives the formula.	Eight groups, three mass profiles per group, loops 1–12: original supplied-equation Kira 1,027.781 s; exact recurrence and shared coefficients 1.922 s. Speedup 534.9×; 99.81% less time. All 120 repeated workflows pass.	One polynomial calculation serves several loop orders. All massive lines share $Q = \sum_h \ell_h$, massless powers equal one, and squared masses are distinct. This control solves supplied Euler and mass identities; it does not time native generation for arbitrary multiloop graphs.
Connected massless two-point bubble trees. Recursive one-loop subintegrals occur in propagator reductions [22].	$J_{\mathcal{T}} = s^{-(A-2L)} \frac{C_{\mathcal{T}}}{C_0} M_{\mathcal{T}}$, $C_{\mathcal{T}} = G(\alpha, c) \prod_{j>0} G(u_j, v_j)$. $s = p^2$, A is the total denominator power, and $M_{\mathcal{T}}$ has unit powers. G is the bubble Gamma ratio in Eq. (88); Eq. (89) defines the powers transferred from each child to its parent. C_0 is the product for the master.	The new method completes 34/34 requests through 13 loops; the preceding application completes 3/34. On those three matched requests, 11.185 → 0.433 s: 25.9× and 96.1% less time.	Integrating leaves successively avoids a large IBP system. Extra occupied propagators can destroy the tree structure. The other 31 requests have no completed control and therefore no finite measured speed factor. This is a gain in both supported coverage and matched time.

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Table 7 continued

Integral and physical context	Final formula and reduction method	Measured control and gain	Applicability and limitation
Connected trees with a massive vacuum root. These are a reducible subset of massive vacuum integrals relevant to zero-momentum expansions [20].	$V_{\mathcal{T}} = (-m^2)^{-(A-P)} \dfrac{C_{\mathcal{T}}^V}{C_0^V} M_{\mathcal{T}}^V$, $P = 2L - 1$. The superscript V denotes the vacuum case. $C_{\mathcal{T}}^V$ multiplies the nonroot bubble factors by the root Gamma ratio in Eq. (92). Appendix N derives the signs and scale power.	Thirty-four matched requests through 13 loops: 7.900 → 5.719 s, or 1.38× and 27.6% less time against the preceding application. A separate cancellation test reduces 34 large batches from 14.305 to 6.731 s, or 52.9% less time.	Early recognition avoids unnecessary symbolic preparation. Cancelling common Gamma factors before expansion helps large batches. In the cancellation test, seven-target batches instead take 2.6% more time, with an interval including equal time. The two savings use different tests and are not multiplied.
Scaleless vacuum sectors. Massless vacuum terms with no external scale arise as removed-line sectors and vanish in dimensional regularization [1].	$I_{\text{scaleless}} = 0$. Scaling and dimensional continuation give the zero identity derived in Appendix O. Recognition avoids generating equations for a sector whose integral vanishes.	Thirteen coupled massless vacuum requests through 13 loops pass the analytic coverage checks. No matched standalone speed factor is available for this subset.	The rule applies only when all dimensionful scales are absent. We report coverage without assigning a speedup to an unsupported native control.

7.1 One-loop tensor results by mass and denominator structure

The aggregate tensor saving in Table 7 contains several physical structures. Table 8 separates them so that the larger group does not hide a poorly performing class. We used rational masses and scalar products, kept d symbolic, and compared every request with the same preceding application. All twenty requests completed in all repetitions. Four mass classes contain only one request each, so their timings describe those cases alone.

Table 8: Exact one-loop numerator reduction by integral type. These gains use angular reduction, power recurrences, and factored coefficient arithmetic. The tensor-order ML model was disabled. Times are sums of three-run request medians.

Integral type	Requests	Before (s)	Exact (s)	Speedup	Less time
Massive bubbles	8	13.833	1.911	7.24×	86.2%
Massive triangles	4	9.020	1.291	6.99×	85.7%
Massive boxes	4	20.998	2.185	9.61×	89.6%
Massless bubble	1	1.409	0.212	6.64×	84.9%
Mixed-mass bubble	1	1.694	0.183	9.28×	89.2%
Mixed-mass triangle	1	1.981	0.270	7.33×	86.4%
Mixed-mass box	1	4.583	0.290	15.78×	93.7%

7.2 Native reductions with the shared structural model

The neural model described in Appendix J used 44 structural features and two hidden layers of 48 components to predict the cost of ten seed choices. We trained it on complete measured cost rather than the length of a constructed reduction path. Table 9 gives its results for every completed physical class in the 110-request grid. The comparison with a global fixed strategy isolates its added request-dependent choice; the native column also includes the benefit or cost of replacing Kira's own choice by our candidate strategies.

For all 43 matched requests, the model takes 57.125 seconds, compared with 63.450 seconds for native choices and 60.094 seconds for the fixed strategy. The corresponding savings are 10.0% and 4.9%. The per-class results include slower cases. In particular, the one-mass vacuum class is slower than native choices even though its point estimate is slightly faster than the fixed strategy. Only two matched requests reach five loops. The remaining 67 requested cases retain the reference failures and coordinate limits shown in Fig. 4.

Table 9: All physical classes with completed shared-model comparisons. Times sum request medians; negative savings mean slower. The last column gives the learned/fixed time-ratio interval from the saved paired resampling. An interval containing one leaves the added gain unresolved. Small classes do not establish population-level transfer.

Integral class and matched loops	Physical context	Cases	Native (s)	Fixed (s)	ML (s)	Saving vs native / fixed	ML/fixed interval
Box with bubble insertions, 2–3 loops	Four-point scattering with bubble subgraphs [1].	4	12.897	7.848	7.696	+40.3% / +1.9%	[0.843, 1.004]
Four-point ladder, 1–2 loops	Four-particle scattering, including planar double boxes [23].	4	8.227	8.756	9.126	−10.9% / −4.2%	[0.938, 1.062]
Nonplanar double box, 2 loops	Crossed two-loop four-point integrals [24].	1	4.235	4.360	4.406	−4.0% / −1.0%	[1.003, 1.040]
One-virtuality box, 1 loop	Four-point amplitudes with one off-shell leg [7].	2	1.885	2.132	2.058	−9.2% / +3.5%	[0.934, 1.084]
Pentagon, 1 loop	Five-parton QCD scattering amplitudes [21].	2	2.548	2.720	2.489	+2.3% / +8.5%	[0.775, 0.951]
Massive two-point, 1–4 loops	Mass-dependent propagator corrections [1].	7	7.428	7.921	7.112	+4.3% / +10.2%	[0.848, 0.953]
Massless two-point, 1–4 loops	Massless propagator corrections and QCD renormalization [22].	7	8.765	7.036	6.949	+20.7% / +1.2%	[0.901, 1.070]
Distinct-mass vacuum, 2–5 loops	Massive vacuum sectors in heavy-quark matching [20].	7	9.388	9.954	8.084	+13.9% / +18.8%	[0.741, 0.890]
One-mass vacuum, 1–5 loops	Massive vacuum sectors in heavy-quark matching [20].	9	8.076	9.367	9.205	−14.0% / +1.7%	[0.945, 1.026]

The final application uses a later, restricted seed-cost tree and fixed arithmetic settings; it does not deploy every historical selector at once. On its last five-request native subgroups, planar double-box time changes from 8.948 to 7.191 seconds and nonplanar double-box time from 14.517 to 11.137 seconds. These are 19.6% and 23.3% savings against the preceding integrated application. The sunrise subgroup instead changes from 7.509 to 8.407 seconds, or 12.0% more time, and the three-loop vacuum subgroup changes from 12.971 to 12.988 seconds. Appendix V gives the shared protocol. These final gains include arithmetic and application changes and do not isolate a new ML effect.

The tables therefore support a conditional use of the method. We can replace reduction by a formula for the proved analytic domains, and we can use a learned strategy when its matched timing control supports it. A connected multiscale integral outside those domains still requires native reduction and a separate performance test.

8 Discussion

Our measurements show that useful learning requires a computational decision whose saved algebra exceeds the prediction and recovery costs. The seed and equation models meet that condition on several stated workloads, and cost-trained selectors improve some matched fixed strategies. The original row model does not. The constructed-data results explain why: a model can reproduce a reference path accurately while taking longer to select each step.

Exact identities provide the largest gain in the restricted-vacuum comparison because they replace many separate solves by a recurrence shared across loop orders. The same mathematical principle motivates the later analytic stage: inspect the propagators, establish that a formula applies, and reduce only the remaining unknown part. Joint target processing provides another substantial gain when requests share a family and kinematics. These are separate sources of acceleration, so their savings cannot all be assigned to ML.

The physical scope has three levels. First, the 17,100-instance constructed collection supplies known-answer targets across a small set of underlying structures. Second, native tests cover selected vacuum, two-point, and scattering families, including some symbolic-kinematic comparisons; the broad shared-model grid completes matched tests through five loops. Third, exact recurrences and formulas cover restricted independent products and connected trees through thirteen loops. A thirteen-loop product or bubble tree has algebraic structure that a general multiscale connected scattering graph may lack.

We also tested finite optimality certificates for small supplied identity pools (Appendix I). Such a certificate depends on the permitted source rows, operations, protected masters, and arithmetic-cost definition. It does not prove a globally shortest IBP reduction or a minimum elapsed time. Likewise, agreement at finite-field samples supports a basis check over the tested ranges, while exact returned coefficients and the analytic derivations provide the stronger checks used for accepted outputs.

Recovery remains a practical limit. A model can remove many equations and still lose time when the reduced system leaves additional masters. Training on complete cost, providing the required basis in the features, and retaining conservative choices reduce this risk without guaranteeing success on every unseen input. The late equation-filter failure and the slower subgroups of the final test remain in the results for this reason.

Table 10 compares the advantages and costs supported by these measurements. The distinction matters when choosing a method for a new calculation. An exact formula needs a proof that the propagators satisfy its assumptions, whereas a learned choice needs evidence that its complete cost is lower than a suitable fixed choice. Native Kira remains the general reduction stage for inputs accepted by the application when the specialized analytic conditions do not hold [4, 5].

Table 10: Advantages, disadvantages, and resulting use of the methods we tested. All performance statements refer to the comparisons reported above.

Method	Advantage	Disadvantage or cost	Consequence for use
Exact formulas and recurrences	Remove equation generation or elimination within a proved domain; give the largest measured savings	Restricted masses, invariants, or graph structure; recognition and output can dominate a short calculation	Check the conditions first, then use the formula for accepted inputs
Learned seed and strategy choices	Reduce complete native time on several matched controls; allow different choices for different targets	Need timed training data; smaller regions can require recovery; transfer varies by physical class	Retain only validated choices and include prediction and recovery in the timer
Learned equation filter	Avoids constructing unused relations; saves 6.7% on its new-index test	A later nonplanar test retries every filtered attempt and is slower	Enable only in a separately supported domain; disabled in the final comparison
Learned row choice	Uses the evolving matrix to rank allowed elimination steps	Original model is 30.5% slower in native trials; compiled policy has an unresolved complete-time benefit	Omit it and retain native row choices
Shared targets and components	Reuses common equations or exact component reductions; joint native requests save 48.3%	Requires compatible family definitions, kinematics, and bases; does not imply a faster isolated request	Share work only when the mathematical inputs match
Fixed validated settings	Avoid model cost and can be faster when a single action suits the domain	Cannot adapt to each new request; a good fixed strategy in one domain can be poor in another	Use as both a deployment option and the matched ML control

Together, these results support a combined procedure whose choice depends on the integral structure and measured cost. We obtained learned savings in specified native reductions, a large cumulative saving for the proved vacuum recurrence, and substantial savings from shared requests. The latest 8.8% update is one part of that evidence. No single number represents all integral types, all loop orders, and all modes of use.

9 Conclusion

We developed and tested a combined approach to speeding up Feynman integral reduction with Kira. We constructed master-based training targets with exact seed and elimination records, trained and tested the original seed and row models, integrated learned choices into native equation generation, and redesigned later models around complete measured reduction cost. Exact coefficient checks and preserved failure records distinguish a useful prediction from a completed reduction.

The learned contribution is supported by specific matched tests. Equation selection saves about 6–7% on its successful native workloads, and the shared structural model saves 4.9% against its fixed control on 43 completed requests. Selective seed expansion and basis conversion give further gains on their stated graph set. Learned row selection is omitted because its complete-time tests do not establish a reliable retained benefit.

The mathematical contribution is larger where the integral structure permits it. Exact recurrences and shared loop dependence reduce the restricted-vacuum benchmark from 1,027.781 to 1.922 seconds, while joint processing of compatible native requests saves 48.3%. Direct formulas and reconstruction give a separate cumulative reduction of 39.1% on 75 mixed requests. The final 93-request comparison adds an 8.8% saving against its preceding application. These measured gains have different controls and must remain separate.

The resulting method uses exact formulas where their assumptions hold, validated learned choices where they reduce cost, and Kira for the remaining algebra. Its analytic tests reach thirteen loops for restricted structures. General connected high-loop scattering, universal ML gains, and globally optimal reduction paths remain unestablished. The report states these limits alongside the methods and gains achieved in this study.

A IBP identities, row operations, and exactness

The starting total-derivative identity follows from integration by parts in a region where the integral converges, followed by dimensional analytic continuation [2, 1]. Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k = \partial/\partial\ell_k$. The product rule gives

$$\partial_k \cdot (v P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} - \sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{11}$$

For $v = \ell_h$, the first factor is $d\delta_{kh}$; for an external momentum it is zero. Completeness of the inverse-propagator coordinates permits us to write $v \cdot \partial_k D_i = u_i + \sum_j v_{ij} D_j$, where u_i and v_{ij} are coefficients fixed by the propagators and kinematics, with the invariant and mass terms included in u_i. Integrating Eq. (11) therefore gives

$$0 = (\partial_k \cdot v) I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma} + \boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma} + \boldsymbol{e}_i - \boldsymbol{e}_j}. \tag{12}$$

The unit vector $\boldsymbol{e}_i$ raises index i by one. This derivation explains how a seed and an operator generate a row of the reduction system.

For the introductory tadpole example, put $D = \ell^2 - x + i0$ and $v = \ell$. Direct differentiation gives

$$\partial_\ell \cdot (\ell D^{-a}) = dD^{-a} - 2a\ell^2 D^{-a-1}. \tag{13}$$

Using $\ell^2 = D + x$ with the infinitesimal prescription understood, then integrating, gives $(d - 2a)T_a - 2axT_{a+1} = 0$. Solving at $a = 1$ proves Eq. (5). This uses the same analytically continued total-derivative identity as Eq. (3).

Let q be a source row and k a pivot column with $A_{qk} \neq 0$. Solving the identity $\sum_j A_{qj} I_j = 0$ for I_k gives $I_k = -\sum_{j \neq k} (A_{qj}/A_{qk}) I_j$. Subtracting A_{rk}/A_{qk} times that zero identity from row r yields

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{14}$$

Every step preserves the represented relation. Repetition until only prescribed master columns remain gives Eq. (6). Removing seeds or equations preserves the validity of the retained identities, but can prevent this elimination from finishing. We therefore require reduction to the prescribed basis and exact agreement with the reference coefficients. Generic kinematics avoid singular pivot denominators; special kinematic limits require their own treatment.

For independent components, write the loop momenta as disjoint groups and the integrand as $\prod_b f_b$. In a common convergence domain, repeated integration gives $I = \prod_b \int f_b$. If the chosen integer powers have no such domain in d, first vary the propagator powers independently and then continue them to the requested values. Analytic continuation preserves the resulting identity [1]. Substituting each exact component reduction therefore gives

$$I = \prod_b \left(\sum_m r_{bm} M_{bm} \right). \tag{15}$$

Our product output retains this factored form. It avoids expanding the Cartesian product of master terms and uses the same representation in all timed methods.

B The seed regions and equation score

For seed $\boldsymbol{\sigma}$, define excess positive power $D = \sum_i \max(\sigma_i - 1, 0)$, numerator rank $R = \sum_i \max(-\sigma_i, 0)$, and positive sector $P = \{i : \sigma_i > 0\}$. Let h be the number of absent physical lines relative to the largest top sector, and let D_T, R_T be the largest target values. The distance from the target numerator rectangles is defined by

$$q(\boldsymbol{\sigma}, \mathcal{T}) = \min_{\boldsymbol{a} \in \mathcal{T}} \sum_{i \notin P} \max\{0, -\sigma_i - \max(0, -a_i)\}. \tag{16}$$

Projection onto the interval from zero to each target numerator power leaves precisely this positive excess. Summing the distances and minimizing over targets gives Eq. (16). We define the region by

$$\begin{aligned} q &\leq \max\{0, \lfloor \theta_0 + \theta_1 h + \theta_2 D + \theta_6 R \rfloor\}, \\ R &\leq \max\{0, \lfloor R_T + \theta_3 - \theta_4 h \rfloor\}, \qquad D \leq \max\{0, \lfloor D_T + \theta_5 \rfloor\}. \end{aligned} \tag{17}$$

These inequalities specify a search family rather than a sufficiency theorem. Native outer bounds, retained sector corners, and admissible target and master seeds also apply. The previously studied sparse choices use

$$\begin{aligned} \boldsymbol{\theta}_{\text{wide}} &= (2.5, 0, 0, 1.25, 0, 1, -0.25), \\ \boldsymbol{\theta}_{\text{tight}} &= (1, 0, 0, 0, 1, 1, 0), \\ \boldsymbol{\theta}_{\text{medium}} &= (1.5, 0, 0, 0.5, 0, 1, 0). \end{aligned} \tag{18}$$

The retained native model uses these three sparse choices. The rejected replacement uses the wide and tight choices and a third action that retains all seeds within the tight procedure's native outer bounds. The native procedure retains its subsector truncation and recovery rules. The model selects one complete procedure before generating equations.

The separately tested equation classifier has seventeen features: four family and request dimensions; three seed counts; two differences from target power maxima; three sector descriptors; exact target membership and minimum index distance; the operator index; and the two target maxima. Positive labels mark equations retained by a complete reference reduction. Sampling at most 2,000 records of each class y in batch c gives each sampled record weight $N_{cy}/(n_{cy}N_c)$, where N_{cy} and n_{cy} are the native and sampled class counts and N_c is the full batch count. Multiplying this weight by n_{cy} preserves the class fraction N_{cy}/N_c. A leaf with weighted totals W_0, W_1 returns $p = W_1/(W_0 + W_1)$, and a fixed threshold retains equations. The decision-tree construction follows Ref. [17]. This empirical score does not certify sufficiency or probability calibration; the native completion check remains necessary.

C Measured-cost learning

Let i label a training request and a an allowed arithmetic procedure. In the latest study, C_{ia} is the measured complete time, including recovery, and $b_i > 0$ is the time for the preceding procedure on the same request. Define the relative training target by

$$y_{ia} = C_{ia}/b_i. \tag{19}$$

There are five actions, so the latest model predicts the vector $\boldsymbol{z}_i = (y_{i1}, \ldots, y_{i5})$. These quantities are training definitions. The three-loop structure uses the preceding procedure by an explicit guard and is excluded from the 36-request fit. The earlier reconstruction study used six actions and an estimated recovery penalty; its separate objective and outcomes remain in the notes.

The ensemble has 32 extremely randomized regression trees [18]. At a terminal branch, or leaf, let S be the set of training requests that reach it and let μ be the predicted value of one output component z. We use the loss $\sum_{i\in S} b_i(z_i - \mu)^2$, which defines the baseline-time weighting. Differentiating gives

$$0 = 2\sum_{i\in S} b_i(\mu - z_i), \qquad \widehat{z}(S) = \frac{\sum_{i\in S} b_i z_i}{\sum_{i\in S} b_i}. \tag{20}$$

Substituting the relative cost from Eq. (19) gives

$$\widehat{y}_a(S) = \frac{\sum_{i\in S} C_{ia}}{\sum_{i\in S} b_i}, \qquad \arg\min_a \widehat{y}_a(S) = \arg\min_a \sum_{i\in S} C_{ia}. \tag{21}$$

The second equality follows because the denominator is positive and independent of the action. Thus an individual leaf ranks actions by their aggregate measured training cost. The ensemble averages its 32 tree predictions. This construction does not establish a minimum cost on unseen requests.

For the fixed action a_0 selected for the request's propagator structure, an alternative a_* is proposed only if

$$\widehat{y}_{a_*} < (1 - \eta)\widehat{y}_{a_0}. \tag{22}$$

The margin η is a defined acceptance parameter. Validation compares $\eta = 0, 0.05, 0.10$, tree depths 2, 4, and 6, and minimum leaf populations 2 and 4. The complete selected policy must cost less than 0.97 times its structure-dependent fixed control and select no failed validation action. The latest policy does not pass this condition. The final application therefore uses full reconstruction with eight evaluation points grouped together for the planar box, nonplanar box, and sunrise structures, and the preceding procedure for the three-loop vacuum structure.

To specify the features, define target excess power $D = \sum_j \max(a_j - 1, 0)$, numerator rank $R = \sum_j \max(-a_j, 0)$, and line count $P_T = \sum_j \mathbf{1}_{a_j>0}$. A line is incident on a loop momentum when that momentum occurs in its inverse propagator. The 29 features are listed in Table 11. Every feature follows from the supplied indices and propagator definitions. The ordering within

Table 11: Inputs to the algebra-cost model. Means and extrema over targets use the complete requested batch.

Positions	Quantities
1–6	Number of loops, external momenta, integral coordinates, physical propagators, symbolic external invariants, and $\log(1 + n_T)$ for n_T targets
7–13	Maximum and mean D, maximum and mean R, minimum and maximum P_T, and number of distinct sectors
14–18	Number of massive coordinates; number of physical lines containing several loops; maximum loop incidence per physical line; maximum physical-line incidence per loop; mean external-momentum incidence per coordinate
19–22	Minimum D, minimum R, mean P_T, and maximum absolute index
23–25	Mean excess power on massive coordinates; mean numerator rank weighted by loop incidence; mean excess power weighted by loop incidence
26–29	Maximum sum of positive powers; maximum number of absent physical lines; maximum numerator and denominator bit lengths among rational masses and scalar products

positions 7–10 is maximum D, maximum R, mean D, mean R. The remaining entries follow the listed order. The bit lengths describe the supplied rational numbers and do not encode a family identifier.

D Physical test inputs

All profiles below define squared quantities in a common fixed unit. The zero-invariant families use shifts $(0, p, q, r)$ and masses $(x, y, 0, 0)$, with

$$\begin{aligned} &p^2 = 0, \quad p \cdot q = -h - 1, \quad p \cdot r = -h - 2, \\ &q^2 = -2h - 7, \quad q \cdot r = -h - 3, \quad r^2 = -3h - 13. \end{aligned} \tag{23}$$

The unequal case has $(x, y) = (h+3, h+11)$, the equal case $(h+3, h+3)$, the mixed case $(h+3, 0)$, and the massless case $(0, 0)$. Targets have the form $(1 + \delta_1, 1 + \delta_2, -b_1, -b_2)$, where $\delta_1 + \delta_2$ is the excess positive power and $b_1 + b_2$ is numerator rank.

Ordinary tensor families have P physical lines, one auxiliary coordinate, shifts $q_1 = 0$, $q_{i+1} = p_i$, and

$$m_i^2 = h+11i^2 \ (i \le P), \quad m_{P+1}^2 = 0, \quad p_i^2 = -(10h+17i), \quad p_i \cdot p_j = -(h+i+2j-2) \ (i < j). \tag{24}$$

Appendix P specifies the current profiles, targets, and separate vacuum grids.

For the native planar box, let p_1, p_2, p_3 be massless, $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, and $2p_2 \cdot p_3 = -s-t$. The seven physical line momenta are

$$(\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \ell_1 - \ell_2), \tag{25}$$

and the auxiliary momenta are $(\ell_1 - p_3, \ell_2 - p_1)$. The nonplanar routing follows Ref. [4]:

$$\begin{aligned} (Q_1, \ldots, Q_9) = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2). \end{aligned} \tag{26}$$

The first seven entries are physical and all masses vanish. Its convention gives $p_i^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$, and $t = (p_1 - p_3)^2$. Expanding the squares gives $2p_1 \cdot p_3 = -t$ and

$2p_2 \cdot p_3 = s + t$. The low-rank profiles use $(s, t) = (-1009 - 12h, -1021 - 14h)$ for the planar graph and $(-113-18h, -127-10h)$ for the nonplanar graph. Appendix Q gives the independently chosen high-rank invariants.

Every high-rank box target is $(1, 1, 1, 1, 1, 1, 1, -u, -v)$ with $u, v \geq 0$ and $1 \leq u + v \leq 4$. The current split uses the profiles and batch counts in Appendix Q. Both graph types occur in all splits, so the test measures new batches and kinematic points within these structures.

E Time ratios and uncertainty

For the final shared-call comparison, let U_{mr} be elapsed time for the entire 93-request collection, method m, and repetition r. We define

$$\widetilde{U}_m = \operatorname{median}_{r=1,2,3} U_{mr}, \qquad \rho_{\text{collection}} = \frac{\widetilde{U}_{\text{final}}}{\widetilde{U}_{\text{preceding}}}, \qquad g = 1 - \rho_{\text{collection}}. \tag{27}$$

The saving g follows by dividing the time difference $\widetilde{U}_{\text{preceding}} - \widetilde{U}_{\text{final}}$ by the preceding time. The observed range is $[\min_r U_{mr}, \max_r U_{mr}]$. It describes the three measurements and is not a statistical confidence interval. Internal per-request timers exclude some initialization and collection overhead, so summing their medians need not give $\widetilde{U}_m$.

Let T_{imr} be complete elapsed time for request i, method m, and repetition r. For the set $\mathcal{C}$ completed by both methods in all repetitions, define

$$S_m = \sum_{i \in \mathcal{C}} \operatorname{median}_{r=1,2,3} T_{imr}, \qquad \rho = \frac{S_{\text{new}}}{S_{\text{control}}}, \qquad \text{fractional saving} = 1 - \rho. \tag{28}$$

These definitions compare identical completed requests. Dividing both times by the control time gives ρ, and subtracting gives the fractional saving. Completion counts remain separate because an unfinished reduction has no verified finite speedup. To estimate variation across requests, we draw $|\mathcal{C}|$ paired request indices with replacement and recompute ρ. We report the central 95% of 5,000 such ratios, using the bootstrap construction of Ref. [19]. This interval describes the sampled request collection and does not establish performance for every integral family.

For the historical compiled-row result, the estimator is the geometric mean of paired ratios, $\rho_G = \exp[n^{-1} \sum_{i=1}^{n} \log(t_i^{\text{learned}} / t_i^{\text{native}})]$, where n is the number of matched units in that study. This is the nth root of their product, obtained by applying the logarithm to that product. It weights relative changes equally, unlike the total-time ratio in Eq. (28); the notes preserve the original grouping and resampling procedure.

F Known-answer targets and the original learning objectives

Let each known integral have coefficients $C_{\boldsymbol{a}m}$ in the chosen master basis. A valid source identity has the form $R = \sum_{\boldsymbol{a}} r_{\boldsymbol{a}} I_{\boldsymbol{a}} = 0$. Adding λR to a target gives

$$\mathcal{T}' = \mathcal{T} + \lambda \sum_{\boldsymbol{a}} r_{\boldsymbol{a}} I_{\boldsymbol{a}} = \mathcal{T}, \qquad C'_m = C_m + \lambda \sum_{\boldsymbol{a}} r_{\boldsymbol{a}} C_{\boldsymbol{a}m} = C_m. \tag{29}$$

This proves preservation of the reference coefficients in Eq. (7). To cancel a present term with coefficient $w_{\boldsymbol{b}}$, choose $\lambda = -w_{\boldsymbol{b}}/r_{\boldsymbol{b}}$ when $r_{\boldsymbol{b}} \neq 0$. An identity that contains only one integral with an unknown reduction instead gives

$$h_0 I_{\boldsymbol{b}} + \sum_j h_j I_{\boldsymbol{a}^{(j)}} = 0 \quad \Longrightarrow \quad C_{\boldsymbol{b}m} = -\frac{1}{h_0} \sum_j h_j C_{\boldsymbol{a}^{(j)}m}, \qquad h_0 \neq 0. \tag{30}$$

Thus some constructions produce a single indexed integral; a general sum of masters followed by replacements produces an integral expression. The recorded source identities distinguish these cases.

The measure and denominator factors in Eq. (1) give mass dimension $Ld - 2\sum_i a_i$. If master m has index sum B_m and the target is assigned index sum A, multiplying it by $(Q^2)^{B_m - A}$ gives common dimension $Ld - 2A$, where Q^2 is a nonzero squared-momentum scale. The construction can therefore use dimensionless rational functions of d and invariant ratios without losing dimensional consistency. Neither this identity nor the recorded path proves a minimum number of operations.

For binary seed label $y \in \{0, 1\}$ and predicted probability p, the Bernoulli likelihood is $p^y(1 - p)^{1-y}$. Taking minus its logarithm gives

$$\ell_{\text{seed}} = -y \log p - (1 - y) \log(1 - p). \tag{31}$$

The implemented loss averages candidate losses within each target and then gives targets equal weight. Each seed score s gives its own sigmoid output $p = 1/(1+e^{-s})$, so several seeds can receive label one. For row scores s_j, define $p_j = e^{s_j} / \sum_k e^{s_k}$ over the allowed actions. The likelihood of recorded action j_* is p_{j_*}, giving $\ell_{\text{row}} = -\log p_{j_*}$. These definitions justify training to reproduce recorded choices. The later cost models train on measured execution time.

If TP counts selected reference seeds, FP counts selected nonreference seeds, and FN counts omitted reference seeds, precision and recall are

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad R = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad F_1 = \frac{2}{P^{-1} + R^{-1}} = \frac{2PR}{P + R}. \tag{32}$$

Complete-set agreement instead counts targets with exactly the recorded set. Sufficiency tests whether the selected equations span a reduction to the required masters. Distinct sufficient sets explain why these measures need not agree. These distinctions also avoid misleading accuracy when most candidate labels are zero [25].

Two later reuse steps follow directly from linearity. If targets reduce through intermediate masters $J_j = \sum_m B_{jm} M_m$, then

$$I_i = \sum_j C_{ij} J_j = \sum_m \left(\sum_j C_{ij} B_{jm} \right) M_m. \tag{33}$$

The conversion must include every intermediate master before the final coefficients are accepted. If two requests share the same family and kinematics, reducing the union of their targets gives one coefficient row for each target; selecting the corresponding rows returns each request's answer. Duplicate targets need only one reduction. These are exact algebraic statements; their time savings remain empirical.

G Exact reuse for the restricted vacuum family

The large cumulative saving in Section 6.3 follows from exact recurrences in a particular Euclidean vacuum family. We derive them here to state both the source of the saving and the restrictions on its use. The family has positive, distinct squared masses $\mu_j = \rho_j \mu$:

$$V_{\boldsymbol{b}}^{(L)} = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{\pi^{d/2}} \frac{1}{\ell_1^2 \dots \ell_L^2} \prod_{j=1}^{R} (Q^2 + \mu_j)^{-b_j}, \qquad Q = \sum_{h=1}^{L} \ell_h. \tag{34}$$

Here b_j is a nonnegative integer, $B = \sum_j b_j \geq 1$, R is the number of massive denominators, μ is a common squared-mass scale and ρ_j is a positive mass ratio. Each massless denominator has power one. All massive denominators share Q, so this family has special algebraic dependencies. Its prescribed masters are $M_j^{(L)} = V_{\boldsymbol{e}_j}^{(L)}$. The comparison uses $R = 2, 3, 4$ and $\mu = 1$ in the selected units.

G.1 Euler and mass partial-fraction identities

We first derive two relations that lower the massive propagator powers. Let f be the integrand without the measure. The simultaneous Euler derivative gives $\sum_h \ell_h \cdot \partial_{\ell_h}(\ell_i^2)^{-1} = -2(\ell_i^2)^{-1}$ for each massless factor. Also, $\sum_h \ell_h \cdot \partial_{\ell_h} Q^2 = 2Q^2$. Therefore

$$\sum_h \partial_{\ell_h} \cdot (\ell_h f) = \left[Ld - 2L - 2\sum_j b_j \frac{Q^2}{Q^2+\mu_j} \right] f. \tag{35}$$

Use $Q^2 = (Q^2+\mu_j) - \mu_j$ and integrate the derivative as in Appendix A. This gives

$$0 = [L(d-2) - 2B] V^{(L)}_{\boldsymbol{b}} + 2\sum_j b_j \mu_j V^{(L)}_{\boldsymbol{b}+\boldsymbol{e}_j}. \tag{36}$$

For two present massive denominators, the integrand identity $(Q^2+\mu_i) - (Q^2+\mu_j) = \mu_i - \mu_j$ separately gives

$$(\mu_i - \mu_j) V^{(L)}_{\boldsymbol{b}} = V^{(L)}_{\boldsymbol{b}-\boldsymbol{e}_i} - V^{(L)}_{\boldsymbol{b}-\boldsymbol{e}_j}. \tag{37}$$

This second relation is an algebraic partial fraction, so it introduces no additional IBP seed.

G.2 A recurrence shared across loop orders

To make the common loop dependence explicit, define $z = L(d-2)$ and $W^{(L)}_{\boldsymbol{b}} = \mu^{B-1} V^{(L)}_{\boldsymbol{b}}$. Multiplication by this power of μ gives W the same mass dimension as the masters. Substituting in Eqs. (36) and (37) removes the common scale. If $\boldsymbol{C}_{\boldsymbol{b}}(z)$ is the vector of master coefficients of $W^{(L)}_{\boldsymbol{b}}$, then

$$\boldsymbol{C}_{\boldsymbol{b}}(z) = \frac{\boldsymbol{C}_{\boldsymbol{b}-\boldsymbol{e}_i}(z) - \boldsymbol{C}_{\boldsymbol{b}-\boldsymbol{e}_j}(z)}{\rho_i - \rho_j}, \qquad b_i, b_j > 0, \tag{38}$$

$$\boldsymbol{C}_{k\boldsymbol{e}_j}(z) = \frac{2(k-1)-z}{2(k-1)\rho_j} \boldsymbol{C}_{(k-1)\boldsymbol{e}_j}(z), \qquad k \geq 2. \tag{39}$$

The initial vector is $\boldsymbol{C}_{\boldsymbol{e}_j} = \boldsymbol{e}_j$. Every recurrence lowers B, so it terminates at the prescribed masters. Its coefficients depend on L only through z. Induction on B shows that they are polynomials in z with rational coefficients at fixed rational ρ_j. One calculation can therefore supply all requested loop orders by substituting $z = L(d-2)$.

G.3 An independent coefficient check

The recurrence result is checked by obtaining the mixed-denominator coefficients through a Taylor partial-fraction expansion. For a dimensionless auxiliary variable x, write

$$\prod_i (x+\rho_i)^{-b_i} = \sum_{j:b_j>0} \sum_{k=1}^{b_j} \frac{a_{jk}}{(x+\rho_j)^k}. \tag{40}$$

At the pole $x = -\rho_j$, set $t = x + \rho_j$. Multiplying by t^{b_j} and expanding the regular factors shows that the principal-part coefficient is

$$a_{jk} = [t^{b_j-k}] \prod_{i\neq j} (\rho_i - \rho_j + t)^{-b_i}. \tag{41}$$

Here $[t^n]$ extracts Taylor coefficient n and factors with $b_i = 0$ are one. Applying Eq. (39) repeatedly to each pole term gives

$$C_{\boldsymbol{b}j}(z) = \sum_{k=1}^{b_j} a_{jk} \prod_{h=1}^{k-1} \frac{2h-z}{2h\rho_j}, \qquad V^{(L)}_{\boldsymbol{b}} = \mu^{1-B} \sum_j C_{\boldsymbol{b}j}(L(d-2)) M^{(L)}_j. \tag{42}$$

An empty product is defined as one. This Taylor calculation checks every returned polynomial coefficient before the substitution for z and does not follow the chosen sequence of pair recurrences. The derivation requires distinct ρ_j, unit powers for the massless propagators, and the shared momentum Q. These assumptions delimit the family to which the result applies.

H Project-wide timing estimators

For method a, batch b, and repetition r, let T_{abr} be the elapsed time and $\widetilde{T}_{ab}$ its median over repetitions. Given control c, the ratio of summed medians and the equally weighted geometric ratio are

$$R_{\mathrm{sum}} = \frac{\sum_b \widetilde{T}_{ab}}{\sum_b \widetilde{T}_{cb}}, \qquad R_{\mathrm{geom}} = \exp\left(\frac{1}{B}\sum_{b=1}^{B} \log \frac{\widetilde{T}_{ab}}{\widetilde{T}_{cb}}\right). \tag{43}$$

The first expression compares total time when every batch is counted once. The second gives equal weight to each batch's relative change. When physical groups contain different numbers of batches, we first average the log ratios within each group and then average those group values. Exponentiating yields the equal-group geometric mean used in the native model studies.

For either declared ratio R, speedup is $1/R$ because reversing the comparison reverses its numerator and denominator. The fraction of time saved is $(T_c - T_a)/T_c = 1 - R$ for the corresponding estimator. Thus the restricted-vacuum summed ratio gives a speedup of 534.9, while its equal-group geometric ratio gives 543.8. Neither number is obtained by multiplying earlier improvements.

The cumulative vacuum experiment uses eight groups, three mass profiles per group, loop orders one through twelve, and five repetitions per method. Maximum total massive powers are 8, 10, 12, and 14, with two groups at each bound. Requested target counts per profile are 16, 32, 48, and 64, capped by available vectors. Its descriptive interval resamples the eight complete groups 10,000 times, keeping the paired methods and repetitions together [19]. The other comparisons retain their original group, request, or repetition resampling units. This describes the uncertainty of each recorded experiment and does not establish a distribution over all Feynman integral families.

I Finite minimum-operation checks

The saved construction supplies a valid reduction path, but a minimum-cost claim requires a defined set of alternatives. We therefore specify the allowed equations, row operations, and cost before checking all paths for two small systems. These bounded examples make precise the distinction between a known path and a certified minimum. Define the massless bubble $B_{a,b} = I_{(a,b)}$ with denominators $\ell^2 + i0$ and $(\ell - p)^2 + i0$, where $p^2 = s \neq 0$. Contracting the derivative with ℓ uses $2\ell \cdot (\ell - p) = \ell^2 + (\ell - p)^2 - s$. Equation (11) then gives

$$0 = (d - 2a - b)B_{a,b} - bB_{a-1,b+1} + bsB_{a,b+1}. \tag{44}$$

Changing variables $\ell \mapsto p - \ell$ exchanges a, b and gives the second operator identity. Integrals with only a massless denominator vanish in dimensional regularization [1]. The two operators at $(1,1)$, the first at $(2,1)$ and the second at $(1,2)$ therefore give

$$\begin{pmatrix} 0 & s & 0 & d-3 \\ s & 0 & 0 & d-3 \\ d-5 & -1 & s & 0 \\ -1 & d-5 & s & 0 \end{pmatrix} \begin{pmatrix} B_{2,1} \\ B_{1,2} \\ B_{2,2} \\ B_{1,1} \end{pmatrix} = 0. \tag{45}$$

The first two rows yield $B_{2,1} = B_{1,2} = (3 - d)B_{1,1}/s$. Substituting in the third yields $B_{2,2} = (d - 6)(d - 3)B_{1,1}/s^2$.

For the massive tadpole $T_a = \int d^d\ell\,(\ell^2 - m^2 + i0)^{-a}/(i\pi^{d/2})$, the same derivative gives

$$0 = dT_a - 2a\int \frac{d^d\ell}{i\pi^{d/2}}\frac{\ell^2}{(\ell^2 - m^2 + i0)^{a+1}} = (d-2a)T_a - 2am^2T_{a+1}. \tag{46}$$

Seeds $a = 1, 2, 3$ reduce T_2, T_3, T_4 to T_1. We keep the master column protected in both systems. An allowed step selects a nonzero pivot in an unprocessed nonmaster column, normalizes its row and eliminates that column from the other rows. The cost counts one division per nonzero nonpivot source entry when normalization is needed, plus one multiplication and one subtraction for each nonpivot source-entry update. It excludes assignments and the internal polynomial operations of a rational-function operation.

Let S be the exact matrix state, $\mathcal{A}(S)$ its allowed actions, $c(S, a)$ the stated step cost and S_a the resulting state. Splitting every complete path into its first action and its remaining path gives the finite recurrence [26]

$$V(S) = \min_{a\in\mathcal{A}(S)}\{c(S,a) + V(S_a)\}, \qquad V(S_{\text{done}}) = 0. \tag{47}$$

Each action removes an unprocessed column. Induction on the number of remaining columns proves the minimum after all actions have been evaluated. Repeating this calculation for every source-row subset of nonmaster rank three also minimizes over initial equation selection.

Our bubble check covers 113 states, 234 transitions and 152 complete paths across five sufficient subsets. Its minimum is seven coefficient operations with subset selection and thirteen with the full pool. The tadpole check covers 24 states, 32 transitions and fifteen paths; its minimum is seven operations with its single sufficient subset. Each pool requires at least three equations. The smallest number of distinct seed labels is two for the bubble and three for the tadpole. Exact replay and six native Kira master-coefficient comparisons agree. These are minima for the displayed identities, protected basis, allowed operations and cost. They do not establish minimum elapsed time or an optimum over all IBP identities.

J Shared structural model, actions, and fitting

Let n be the number of physical propagators and E the number of independent external momenta. Let B_{ih} be one when loop momentum ℓ_h appears in propagator momentum q_i, and zero otherwise. Table 12 groups the 27 base inputs. Means run over the target batch unless another set is specified. A weighted incidence multiplies the relevant numerator power or extra denominator power by the corresponding row sum of B before summing over propagators.

We add eight ratios: N/n, n/L, massive-entry fraction, coupled-physical-entry fraction, maximum dots divided by n, maximum rank divided by $N - n$, maximum physical loop incidence divided by L, and maximum physical loop degree divided by n. Each denominator is replaced by at least one. Six further features are the mean fractions of indices equal to $-2, -1, 0, 1, 2, 3$. The final three are $\log[1 + L(L + E)]$, $\log(1 + n)(1 + D_T)$, and $\log[1 + \max(1, N - n)](1 + R_T)$. These definitions give 44 features. Permuting loop names or entries within the physical and auxiliary groups preserves the pooled counts when the indices are permuted consistently. A change of momentum routing need not preserve them, and identical features need not identify identical reduction problems.

Let (r_0, s_0, d_0) be the target maxima of positive-power sum, numerator rank, and extra positive power. Table 13 defines ten candidate actions. Bounds are nonnegative and r is at least the number t_{top} of positive entries in the largest top sector. Kira's subsector restriction uses [5]

$$s \le \max\{1, t - l + 1, s_{\max,\text{sector}}\}, \tag{48}$$

where t is that sector's positive-index count, l is the chosen parameter, and the last term is an explicitly assigned sector limit.

Table 12: The base structural inputs. Each quantity is computed from the family definition and target indices before Kira generates equations.

Group	Quantities	Number
Family dimensions	L, E, N, n; number of symbolic invariants	5
Propagator structure	Massive-entry count; coupled-physical-entry count; maximum physical row and column sums of B; mean external-momentum incidence over all entries	5
Target batch	Log of one plus target count; number of target sectors	2
Power ranges	Minimum, mean, and maximum extra positive powers and numerator rank	6
Positive indices	Minimum, mean, and maximum number of positive indices	3
Further index limits	Largest absolute index; maximum positive-power sum; maximum number of absent physical denominators	3
Weighted powers	Mean dots on massive entries; mean numerator-weighted and dot-weighted loop incidence	3

Table 13: The shared model's candidate seed configurations. Changes are relative to the target maxima. A dash disables the additional subsector restriction.

Action	Δr	Δs	Δd_0	l
Native	0	0	0	—
Numerator expansion	0	1	0	—
Tight	-1	-1	-1	—
Reduced dots	-1	0	-1	—
Reduced numerator rank	0	-1	0	—
Expanded	1	1	1	—
Truncated native	0	0	0	$t_{\text{top}} - s_0$
Truncated expansion	0	1	0	$t_{\text{top}} - s_0$
Truncated reduced dots	-1	0	-1	$t_{\text{top}} - s_0$
Relaxed truncation	0	0	0	$t_{\text{top}} - s_0 - 1$

For request c and action a, define the cost label $y_{ca} = \log(T_{ca}/T_{c0})$, with native action 0. Because the denominator is positive and independent of a, and the logarithm is increasing, minimizing the unmodified label selects the action with the least recorded time. Training clips this label to $[-4, 5]$ after assigning failures at least 180 seconds or three times the measured attempt cost, whichever is larger. Clipping limits extreme labels and can merge their order; the validation comparison therefore uses the recorded costs directly.

The neural regression model uses composed affine maps and componentwise activations [27]. The selected network standardizes each feature by its training mean and standard deviation, with a minimum scale of 10^{-6}. With standardized vector z, its outputs are defined by

$$h_1 = \mathrm{ReLU}(W_1^{\mathsf{T}} z + b_1), \quad h_2 = \mathrm{ReLU}(W_2^{\mathsf{T}} h_1 + b_2), \quad \widehat{y} = W_3^{\mathsf{T}} h_2 + b_3, \tag{49}$$

where $\mathrm{ReLU}(u) = \max(0, u)$ acts on each component. Both hidden vectors have 48 components and the output has ten components, one per action. We fit squared prediction error with a squared-weight penalty of strength 0.1 using the regression implementation of Ref. [27]. For n_{tr} training requests, the corresponding objective is

$$\mathcal{L} = \frac{1}{2n_{\mathrm{tr}}} \sum_{c,a} (\widehat{y}_{ca} - y_{ca})^2 + \frac{0.1}{2n_{\mathrm{tr}}} \sum_j \|W_j\|_F^2. \tag{50}$$

To motivate the first term, assign a formal residual $e_{ca} = \widehat{y}_{ca} - y_{ca}$ the density $p(e) = (2\pi)^{-1/2} \exp(-e^2/2)$. Then

$$-\frac{1}{n_{\mathrm{tr}}} \log \prod_{c,a} p(e_{ca}) = \frac{1}{2n_{\mathrm{tr}}} \sum_{c,a} e_{ca}^2 + \frac{10}{2} \log(2\pi).$$

Removing the constant and adding the stated quadratic penalty gives Eq. (50). This construction motivates a regression objective; the study does not infer a Gaussian distribution of timing errors.

The candidate set also contains trees, random forests, and extremely randomized forests at depths 2, 4, and 7, with at least three samples per leaf; forests contain 48 trees [17, 28, 18]. Two ridge fits use regularization strengths 1 and 10, and the neural candidates have two hidden layers of either 24 or 48 components. Neural fitting uses L-BFGS with at most 500 iterations [27]. The initial implementation exposed an inconsistent scaling between the installed multi-output loss and its gradient. We corrected the loss to match the summed-output gradient, checked it by finite differences, and refitted using the same training and validation inputs. The initial comparison remains in the notes; the corrected model uses a new reserved profile. Candidate convergence warnings remain recorded, and no fit certifies a global optimum.

For every candidate, five-fold cross-validation withholds whole ordered graph structures, with feature scaling fitted within each training fold. The separate validation set selects a single fixed action. We compare that action and native bounds as global anchors, with margins $0, 0.05, 0.12, 0.25$. If b is the anchor and $a_* = \arg\min_a \widehat{y}_a$, the policy changes to a_* only when $\widehat{y}_{a_*} < \widehat{y}_b - \eta$. Exponentiation gives the predicted ratio condition $\exp(\widehat{y}_{a_*} - \widehat{y}_b) < \exp(-\eta)$. We select the candidate minimizing the worse of its grouped-fold and separate-validation ratios to native cost. This rule is fixed before the reserved test. A score below 0.97 permits default deployment; it is evidence from validation, not a guarantee for future inputs.

The selected candidate is the 48-component network, with action 7 from Table 13 (numbered from zero) as anchor and margin 0.05. Training uses only historical training records after the physical holdouts. Their historical timers differ in whether initial imports are included; all new reserved controls use the same complete fresh-process timer. The training record contains every source and model candidate. A failed new seed attempt retries native and expanded bounds with duplicate configurations removed. The shared model never uses target coefficients or measured test time as an input.

K Physical inputs and the loop-order test

The following equations define the physical inputs for our benchmark families. All denominators follow Eq. (1). Let $Q = \sum_{i=1}^{L} \ell_i$. For $L \geq 2$, the vacuum families contain

$$q_i = \ell_i \quad (1 \leq i \leq L), \qquad q_{L+1} = Q. \tag{51}$$

The one-mass case gives only the final entry a nonzero mass; the massive case assigns distinct nonzero masses to all physical entries. The one-loop vacuum is a single massive tadpole. The two-point families replace the final momentum by $p - Q$, with either all zero masses or distinct nonzero masses. Their single independent external invariant is p^2.

For the four-point families, choose massless external momenta with $p_1 + p_2 + p_3 + p_4 = 0$, and write $P = p_1 + p_2$, $U = p_1 + p_2 + p_3$, $s = P^2$, $t = (p_2 + p_3)^2$. Expanding these squares gives $2p_1 \cdot p_2 = s$, $2p_2 \cdot p_3 = t$, and $2p_1 \cdot p_3 = -s - t$. The physical momenta of the L-loop ladder are

$$\{\ell_i, \ell_i - P : 1 \leq i \leq L\} \cup \{\ell_i - \ell_{i+1} : 1 \leq i < L\} \cup \{\ell_1 - p_1, \ell_L - U\}. \tag{52}$$

We order the loop momenta first, the P-shifted momenta next, then adjacent differences, and finally the two end entries. The box with bubble subgraphs has

$$(q_1, q_2, q_3, q_4) = (\ell_1, \ell_1 - p_1, \ell_1 - P, \ell_1 - U), \qquad (q_{2j+1}, q_{2j+2}) = (\ell_j, \ell_j - \ell_1), \quad 2 \leq j \leq L. \tag{53}$$

All these four-point propagators are massless. This box family has bubble subgraphs and is a restricted connected scattering class.

The additional massless nonplanar box uses the routing of Ref. [4]:

$$\begin{aligned}(q_1, \ldots, q_9) = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2).\end{aligned} \tag{54}$$

Its first seven entries are physical. Here the outgoing-momentum convention gives $p_i^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$, and $t = (p_1 - p_3)^2$. Expansion gives $2p_1 \cdot p_3 = -t$ and $2p_2 \cdot p_3 = s + t$. The pentagon uses five massless external momenta with zero sum and physical momenta ℓ, $\ell - p_1$, $\ell - p_1 - p_2$, $\ell - p_1 - p_2 - p_3$, and $\ell - p_1 - p_2 - p_3 - p_4$. Adjacent invariants are $s_{ij} = (p_i + p_j)^2$. Squaring complementary momentum sums gives, for example, $2p_1 \cdot p_3 = s_{45} - s_{12} - s_{23}$. The box with one virtual external leg uses $(\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_3)$, with $p_i^2 = 0$ for $i = 1, 2, 3$ and $v = (p_1 + p_2 - p_3)^2$; its remaining product is $2p_2 \cdot p_3 = s + t - v$.

There are L loop squares, $L(L-1)/2$ products of distinct loop momenta, and LE loop–external products. Adding these counts gives the complete coordinate number

$$N = \frac{L(L+1)}{2} + LE. \tag{55}$$

This is the representation used by Kira [5]. We append squared momentum differences when their scalar-product coefficient vectors increase the exact rank, until all N coordinates are independent. The candidate order follows loop momenta and then external momenta. Auxiliary indices are nonpositive. The tested interface admits at most 31 coordinates; this is an implementation limit, not an upper bound on the mathematics of IBP. It permits the defined vacuum inputs through seven loops, two-point inputs through six, and four-point inputs through five. Completion remains a separate test.

For the final fresh profile $h = 1213$, choose $m^2_{\text{one}} = 2h+17$, $m_i^2 = 2h+19+6(i-1)$, $p^2 = -2h-29$, $s = -1009-12h$, and $t = -1021-14h$, using only the variables required by each of the six recursive families. All values use a common squared-momentum unit. Additional scattering profiles follow the explicit assignments in Table 14. The independent reference increases the target's positive-power, numerator, and dot limits by one; the check increases the numerator allowance once more and changes integral ordering. Every checked reference and every failure is retained.

Table 14: Additional scattering kinematics at profile $h = 1201$ for the initial shared-model trial and $h = 1213$ for the corrected-model test. These expressions define the chosen inputs in a common squared-momentum unit.

Family	Invariants
Nonplanar box	$s = -113 - 18h$, $t = -127 - 10h$
Box with virtual leg	Same s, t; $v = -149 - 12h$
Pentagon	$(s_{12}, s_{23}, s_{34}, s_{45}, s_{51}) = (-11, -13, -17, -19, -23) - 20h(1, 1, 1, 1, 1)$

Let $a_i^{(0)} = 1$ on the n physical entries and zero otherwise. For $j = 0, \ldots, \min(n, 3) - 1$, set $i_j = 1 + [(h + j) \bmod n]$ and, when $N > n$, $k_j = n + 1 + [(h + j) \bmod (N - n)]$. The dotted batch uses $\boldsymbol{a}^{(0)} + 2\boldsymbol{e}_{i_j}$; the mixed batch uses $\boldsymbol{a}^{(0)} + \boldsymbol{e}_{i_j} - \boldsymbol{e}_{k_j}$. Without an auxiliary entry, the latter has only the extra physical power. The three additional structures use the earlier generator, which puts a second physical dot on the next entry when no auxiliary entry exists. The notes give its full historical target definitions. These are indexed Feynman integrals whose reductions come from exact Kira calculations.

For the retained two-loop sparse-seed test, the planar box has physical momenta $(\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \ell_1 - \ell_2)$ and auxiliary momenta $(\ell_1 - p_3, \ell_2 - p_1)$. Its convention is $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, and $2p_2 \cdot p_3 = -s - t$. The nonplanar family is Eq. (54). Both use unit physical powers and numerator ranks two through four on one or two auxiliary entries. The equation-transfer comparison also includes the three-loop vacuum with momenta $(\ell_1, \ell_2, \ell_3, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3)$ and the massive two-point ladder $(\ell_1, \ell_1 - p, \ell_2, \ell_2 - p, \ell_3, \ell_3 - p, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3)$, whose final entry is auxiliary. For the three-loop vacuum the squared masses are $(1, 1, 1, 13, 13, 53 + 2h)$; for the two-point ladder they are $(1, 0, 0, 0, 0, 0, 0, 0)$ on the physical entries, with $p^2 = -1009 - 12h$. The earlier sparse-seed fit uses profiles $h = 831, 832$ for training, $841, 842$ for validation, and $851, 852$ for the 20-request test. Its planar invariants are $(-1009 - 12h, -1021 - 14h)$ and its nonplanar invariants are $(-113 - 18h, -127 - 10h)$. A single-direction numerator carries rank two, three, or four; a distributed numerator splits rank three or four into its ceiling and floor halves on the two auxiliary entries. The equation-filter training uses profiles $511, 512$, validation uses 513, and its new-index comparison uses $605, 606$. The notes preserve the complete finite target enumeration and source records for those earlier experiments.

L Exact reduction of polynomial numerators

L.1 Resolving the loop momentum

One-loop tensor reduction expresses loop-momentum numerators through scalar integrals. Established methods include tensor decompositions and generating functions for arbitrary numerator and denominator powers [7, 8]. We derive the particular identities used in our extension here. The algebra uses rational kinematic values and keeps d symbolic. It applies to a one-loop sector and to independent products of such sectors. Applying it inside a coupled graph would require coefficients that depend on the other loop momenta, which this implementation does not support.

Write $D_i = (\ell - q_i)^2 - m_i^2$ and let $S = \{i : a_i > 0\}$ contain the denominator positions. The q_i in this appendix are fixed external shifts; the infinitesimal prescription is understood. Choose $r \in S$, set $k = \ell - q_r$, and define $R_i = q_i - q_r$. The vectors R_i with $i \in S \setminus \{r\}$ span the external directions seen by the denominators. Their Gram matrix and their contractions with k are

$$G_{ij} = R_i \cdot R_j, \qquad u_i = k \cdot R_i = \frac{D_r - D_i + m_r^2 - m_i^2 + R_i^2}{2}. \tag{56}$$

The second equality follows by expanding $D_i = (k - R_i)^2 - m_i^2$ and subtracting it from $D_r = k^2 - m_r^2$. Assume that G is invertible, and split k into its projection onto this span and its perpendicular part:

$$k_\parallel^\mu = \sum_{i,j \in S\setminus\{r\}} R_i^\mu (G^{-1})_{ij} u_j, \qquad k_\perp = k - k_\parallel. \tag{57}$$

Contracting with each R_i gives $k_\perp \cdot R_i = 0$. Contracting the parallel part with itself gives $k_\parallel^2 = u^T G^{-1} u$, so

$$T(D) := k_\perp^2 = D_r + m_r^2 - u^T G^{-1} u. \tag{58}$$

Empty matrices and sums give zero when S has one element.

An inverse propagator D_j with $a_j < 0$ appears in the numerator and can be resolved into the same components. Define $v_{ji} = R_j \cdot R_i$, $R_{j,\perp} = R_j - \sum_{i,h} R_i (G^{-1})_{ih} v_{jh}$, and $z_j = k_\perp \cdot R_{j,\perp}$. Direct substitution gives

$$D_j = B_j(D) - 2z_j, \tag{59}$$

$$B_j(D) = D_r + m_r^2 + R_j^2 - m_j^2 - 2 v_j^T G^{-1} u, \tag{60}$$

$$H_{jh} := R_{j,\perp} \cdot R_{h,\perp} = R_j \cdot R_h - v_j^T G^{-1} v_h. \tag{61}$$

Thus the remaining task is to average products of the z_j over perpendicular directions. All positive-power denominators depend on $k_\perp$ only through $T(D)$, so this angular average leaves the denominator unchanged.

L.2 Angular moments and the finite numerator expansion

Let $\nu = d - |S| + 1$ be the dimension perpendicular to the denominator span. Rotational symmetry makes odd angular moments vanish. For $2h$ perpendicular components, the invariant symmetric tensor is the sum of all pairings of metric tensors, multiplied by a coefficient $C_h T^h$. Tracing two indices gives

$$C_h(\nu + 2h - 2) = C_{h-1}, \qquad C_0 = 1, \qquad C_h = \frac{1}{\prod_{b=0}^{h-1}(\nu + 2b)}. \tag{62}$$

The trace factor contains ν from the pairing of the traced indices and $2h - 2$ from their pairings with the other indices. Contracting the tensor with the external perpendicular vectors therefore gives

$$\left\langle \prod_{t=1}^{2h} z_{j_t} \right\rangle = \frac{T(D)^h}{\prod_{b=0}^{h-1}(\nu + 2b)} \sum_{\mathcal{P}} \prod_{(s,t)\in\mathcal{P}} H_{j_s j_t}. \tag{63}$$

Here $\mathcal{P}$ runs over the distinct pairings of the $2h$ positions, including their multiplicities when an index repeats. In particular,

$$\langle z_i z_j \rangle = \frac{T H_{ij}}{\nu}, \qquad \langle z_i z_j z_k z_l \rangle = \frac{T^2 (H_{ij}H_{kl} + H_{ik}H_{jl} + H_{il}H_{jk})}{\nu(\nu+2)}. \tag{64}$$

One can first derive these relations after a Wick rotation in an integer dimension, where the angular average is well defined, and then continue the rational tensor coefficients in d. This uses the dimensional-regularization convention of Eq. (1) [1]. Poles in d remain exact rational poles.

For $b_j = -a_j > 0$, expand $\prod_{j:a_j<0}(B_j - 2z_j)^{b_j}$ and insert Eq. (63). The result is a finite polynomial in the positive-sector denominators:

$$\left\langle \prod_{j:a_j<0} D_j^{b_j} \right\rangle = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) \prod_{i\in S} D_i^{t_i}, \qquad I_{\boldsymbol{a}} = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) I_{\boldsymbol{a}_S - \boldsymbol{t}}. \tag{65}$$

The vector $\boldsymbol{a}_S$ equals $\boldsymbol{a}$ on S and vanishes elsewhere; the t_i are nonnegative and vanish outside S. The coefficients also depend rationally on the fixed kinematics. Each power $D_i^{t_i}$ cancels

the corresponding denominator power. If a cancellation creates a new negative index, that line leaves the positive sector. Repeating the angular step therefore decreases the positive-sector size whenever a numerator remains. A sector with no denominators is scaleless and vanishes in dimensional regularization [1]. The process terminates in scalar sectors with nonnegative indices.

L.3 Lowering powers before removing the numerator

A second exact route starts with IBP. Define $A = \sum_j a_j$ and $Y_{ij} = m_i^2 + m_j^2 - (q_i - q_j)^2$. Expanding the total derivative with vector $\ell - q_i$ gives

$$0 = dI_{\boldsymbol{a}} - \sum_j a_j \int \frac{d^d\ell}{i\pi^{d/2}} \frac{2(\ell - q_i)\cdot(\ell - q_j)}{D_j} \prod_h D_h^{-a_h}, \tag{66}$$

$$2(\ell - q_i)\cdot(\ell - q_j) = D_i + D_j + Y_{ij}, \tag{67}$$

$$0 = (d - A)I_{\boldsymbol{a}} - \sum_j a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_j a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{68}$$

Here $\boldsymbol{e}_j$ is the unit index vector. This derivation holds for signed integer indices, so it includes numerator terms. Let $N_- = \{j : a_j < 0\}$. Moving the negative-index terms to the right gives, for $i \in S$,

$$\sum_{j\in S} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j} = F_i(\boldsymbol{a}), \tag{69}$$

$$F_i(\boldsymbol{a}) = (d - A)I_{\boldsymbol{a}} - \sum_{j:a_j\neq 0} a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_{j\in N_-} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{70}$$

For target $\boldsymbol{b}$ with $b_k > 1$, choose $\boldsymbol{a} = \boldsymbol{b} - \boldsymbol{e}_k$. If Y_S is invertible, multiply this system by its inverse to obtain

$$I_{\boldsymbol{b}} = \frac{1}{a_k} \sum_{i\in S} (Y_S^{-1})_{ki} F_i(\boldsymbol{a}). \tag{71}$$

Every integral on the right has a smaller sum of positive indices than $\boldsymbol{b}$, and the numerator rank does not increase. When the positive powers reach one, the angular step removes the remaining numerator. With no negative indices, Eq. (71) reduces to the scalar Cayley recurrence derived earlier. These identities give a finite reduction to unit-power scalar sectors. They produce a spanning set; special kinematics can introduce further relations between its elements.

L.4 Keeping the dimension poles in factored form

The preceding identities restrict the form of every dimension-dependent denominator. The inverse Gram and Cayley matrices have rational entries at the chosen kinematic point, and Eq. (71) introduces only polynomial dependence on d. Equation (63) introduces products of linear factors $d + j$, where j is an integer. By induction through the recurrence, each coefficient therefore has the form

$$r(d) = \frac{P(d)}{\prod_j (d + j)^{n_j}}, \qquad P(d) \in \mathbb{Q}[d], \quad n_j \in \mathbb{Z}_{\geq 0}. \tag{72}$$

The finite products contain only factors introduced by the angular moments. For addition, take Q to be the least common multiple of two such denominators Q_1 and Q_2. Multiplying each summand by its missing factors gives

$$\frac{P_1}{Q_1} + \frac{P_2}{Q_2} = \frac{P_1(Q/Q_1) + P_2(Q/Q_2)}{Q}. \tag{73}$$

The exponent of $d+j$ in Q is the larger of its two exponents. Multiplication adds these exponents. A factor $d+j$ cancels exactly when $P(-j) = 0$, because division with remainder gives $P(d) = (d+$

$j)\widetilde{P}(d) + P(-j)$. We repeat this cancellation for each retained factor. These operations preserve the exact rational function and require only rational polynomial coefficients and integer factor multiplicities. They avoid general rational-function simplification in the new path. The earlier arithmetic checks compare these coefficients with a general rational-function implementation and native Kira reductions. The current application retains the fixed numerator-first recurrence; its new learned decision concerns native seed regions.

M Exact reduction at a zero two-point invariant

A singular Gram matrix prevents the projection used by the generic tensor recurrence. For a two-point sector with $p^2 = 0$, Feynman parameters give a direct reduction to tadpoles. Degenerate one-loop kinematics require such separate identities [7]. We derive the identity used in this study below. It applies to a polynomial numerator and two positive denominator powers; it does not assume that the two masses differ.

M.1 Scalar integrals and mass derivatives

Let x and y denote squared masses, and define

$$T(x) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{\ell^2 - x + i0}, \qquad B_{ab}(x,y) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{(\ell^2 - x + i0)^a [(\ell - p)^2 - y + i0]^b}\bigg|_{p^2=0}. \tag{74}$$

Here a, b are positive integers. Dimensional regularization sets $T(0) = 0$. With the normalization in Eq. (1), Wick rotation and the radial beta integral give [1]

$$T(x) = -\Gamma(1 - d/2)(x - i0)^{d/2-1}. \tag{75}$$

For $a = b = 1$, combine the denominators with a parameter z and shift $\ell = k + (1 - z)p$. The combined squared mass is $M^2(z) = zx + (1 - z)y$, because $p^2 = 0$. Integrating over k and then z gives

$$\begin{aligned} B_{11}(x,y) &= \Gamma(2 - d/2) \int_0^1 [zx + (1 - z)y - i0]^{d/2-2}\, dz \\ &= \frac{T(x) - T(y)}{x - y}, \qquad x \neq y. \end{aligned} \tag{76}$$

The second equality follows from the elementary antiderivative of a power and $\Gamma(2 - d/2) = (1 - d/2)\Gamma(1 - d/2)$. Differentiating a propagator with respect to its squared mass raises its power. Therefore

$$B_{ab}(x,y) = \frac{\partial_x^{a-1}\partial_y^{b-1} B_{11}(x,y)}{(a-1)!(b-1)!}. \tag{77}$$

This operation preserves a rational combination of $T(x)$ and $T(y)$ because $\partial_x T(x) = (d/2 - 1)T(x)/x$. We take the derivatives before imposing relations between the masses. If $x = y \neq 0$, the combined mass is independent of z, and the beta integral instead gives

$$B_{ab}(x,x) = T_{a+b}(x), \qquad T_n(x) = \frac{\prod_{j=1}^{n-1}(d/2 - j)}{(n-1)!\, x^{n-1}} T(x). \tag{78}$$

The formula for T_n also follows by differentiating Eq. (75). These identities support independent symbolic squared masses as well as rational values. A relation involving distinct symbolic masses is a generic rational identity; equal masses use Eq. (78) directly.

M.2 Polynomial numerators

Let $P(\ell)$ be a polynomial numerator. Combining the two denominators gives

$$B_{ab}[P] = \frac{\Gamma(A)}{\Gamma(a)\Gamma(b)} \int_0^1 dz\, z^{a-1}(1-z)^{b-1} \int \frac{d^dk}{i\pi^{d/2}} \frac{P(k+(1-z)p)}{[k^2 - M^2(z) + i0]^A}, \quad A = a+b. \tag{79}$$

For an auxiliary inverse propagator $(\ell - q_j)^2 - m_j^2$, put $w_j = (1-z)p - q_j$. Its numerator factor becomes $k^2 + 2k\cdot w_j + w_j^2 - m_j^2$. Both $w_i \cdot w_j$ and $w_j^2 - m_j^2$ are polynomials of degree at most one in z when $p^2 = 0$.

Rotational covariance gives zero for odd moments of k. Tracing two indices of an even moment determines its normalization recursively: if its pairing coefficient is C_h, then $C_h(d+2h-2) = C_{h-1}$ and $C_0 = 1$. Consequently,

$$\left\langle \prod_{j=1}^{2h} (k\cdot w_j) \right\rangle = \frac{(k^2)^h}{\prod_{r=0}^{h-1}(d+2r)} \sum_{\mathcal{P}} \prod_{(i,j)\in\mathcal{P}} w_i \cdot w_j, \tag{80}$$

where $\mathcal{P}$ runs over pairings of the $2h$ positions. The trace recurrence follows because one pairing contributes d and the remaining contractions contribute $2h-2$. This also derives the full-dimensional form of the angular identity used in Appendix L.

After the finite numerator expansion and angular average, define $c_{ht}(d)$ by

$$\langle P(k+(1-z)p)\rangle = \sum_{h,t} c_{ht}(d)(k^2)^h z^t. \tag{81}$$

These coefficients also depend on the fixed external scalar products and numerator masses. They contain no loop momentum. To integrate each term, Wick rotate the radial integral, set $u = k_E^2/M^2$, and use $\int_0^\infty u^{\alpha-1}(1+u)^{-A}du = \Gamma(\alpha)\Gamma(A-\alpha)/\Gamma(A)$ with $\alpha = d/2 + h$. To obtain this beta identity, multiply the defining Gamma integrals for $\Gamma(\alpha)$ and $\Gamma(A-\alpha)$, and set their variables to rz and $r(1-z)$. The r integral is $\Gamma(A)$; the remaining integral is converted by $u = z/(1-z)$ to the stated expression. This gives

$$J_{Ah}(M^2) = \int \frac{d^dk}{i\pi^{d/2}} \frac{(k^2)^h}{(k^2 - M^2 + i0)^A} = (-1)^{A-h} \frac{\Gamma(d/2+h)\Gamma(A-h-d/2)}{\Gamma(d/2)\Gamma(A)} (M^2 - i0)^{d/2+h-A}. \tag{82}$$

The initial convergence region defines the integrals; analytic continuation gives the dimensional-regularization identity outside that region [1].

For $x \neq y$, set $\Delta = x - y$, $\rho = h - A + 1$, and $\lambda = d/2 - 1$. Expand $(1-z)^{b-1}$ and then substitute $u = y + \Delta z$. For an integer $k \geq 0$,

$$\int_0^1 z^k (y + \Delta z)^{\lambda+\rho}\, dz = \frac{1}{\Delta^{k+1}} \sum_{j=0}^{k} \binom{k}{j} (-y)^{k-j} \frac{x^{\lambda+\rho+j+1} - y^{\lambda+\rho+j+1}}{\lambda+\rho+j+1}. \tag{83}$$

This follows by expanding $(u-y)^k$ and integrating each power of u. Dividing by the tadpole normalization in Eq. (75), define

$$R_h = \frac{(-1)^{A-h+1}(d/2)_h (1-d/2)_{A-h-1}}{(a-1)!(b-1)!}, \qquad (z)_n = \frac{\Gamma(z+n)}{\Gamma(z)}. \tag{84}$$

For a negative integer n, the last ratio means the inverse finite product $1/\prod_{r=1}^{-n}(z-r)$. Equations (79)–(83) now give the explicit rational reduction

$$\begin{aligned} B_{ab}[P] = \sum_{h,t} c_{ht}(d) R_h \sum_{v=0}^{b-1} (-1)^v \binom{b-1}{v} \\ \times \sum_{j=0}^{k} \binom{k}{j} \frac{(-y)^{k-j}}{\Delta^{k+1}} \frac{x^{\rho+j+1}T(x) - y^{\rho+j+1}T(y)}{\lambda+\rho+j+1}, \\ k = a - 1 + t + v. \end{aligned} \tag{85}$$

If one mass is zero, its endpoint vanishes in a convergence region and hence by analytic continuation; the implementation omits that scaleless endpoint before evaluating powers. If both masses vanish, the sector is scaleless. For equal nonzero masses, use the beta integral directly:

$$B_{ab}[P] = \sum_{h,t} c_{ht}(d) R_h \, \frac{\Gamma(a+t)\Gamma(b)}{\Gamma(A+t)} x^{h-A+1} T(x). \tag{86}$$

Thus the equal-mass rule has no division by $x - y$. A sector with one positive denominator uses Eq. (82) and the same angular expansion without a Feynman parameter.

At rational kinematics, the coefficients have polynomial numerators in d and products of integer-shift linear factors $d + j$ in their denominators. The finite products in Eqs. (80) and (84), and the linear denominator in Eq. (83), prove this statement. We therefore use the exact factored arithmetic of Appendix L. The implemented tensor domain has at most five coordinates, numerator rank at most four, and at most four extra positive powers. Symbolic masses currently use the scalar derivative identity. Inputs outside these conditions continue through the existing exact rules and Kira.

N Connected one-scale tree reductions

We derive the previously retained tree rules to state precisely which high-loop structures they cover. Let $\mathcal{T}$ be a rooted tree with loop momenta $\ell_0, \dots, \ell_{L-1}$ and parent $\pi(j)$ for every nonroot vertex. A massless two-point family is

$$J_{\mathcal{T}} = \int \prod_{j=0}^{L-1} \frac{d^d \ell_j}{i\pi^{d/2}} \frac{1}{(\ell_0^2)^{a_0}[(\ell_0 - p)^2]^c} \prod_{j=1}^{L-1} \frac{1}{(\ell_j^2)^{a_j}[(\ell_j - \ell_{\pi(j)})^2]^{v_j}}, \qquad p^2 = s \neq 0. \tag{87}$$

The massive vacuum version replaces the two root factors by $(\ell_0^2 - m^2)^{-c}$ and has no external momentum. Each edge power v_j, root power c, and leaf self power a_j is positive. Internal self powers can be nonpositive. Auxiliary powers vanish. The implementation admits $L \leq 13$ and absolute physical powers at most sixteen. The derivation itself applies to finite trees with meromorphically defined integrals.

Temporarily use negative massless denominators. Feynman parametrization of a massless bubble, translation of its loop momentum, and a radial beta integral give

$$\begin{aligned} G(u,v) &= \frac{\Gamma(u+v-d/2)}{\Gamma(u)\Gamma(v)} \int_0^1 z^{d/2-v-1}(1-z)^{d/2-u-1}\, dz \\ &= \frac{\Gamma(u+v-d/2)\Gamma(d/2-u)\Gamma(d/2-v)}{\Gamma(u)\Gamma(v)\Gamma(d-u-v)}. \end{aligned} \tag{88}$$

The remaining scale factor is $(-P^2 - i0)^{d/2-u-v}$ for bubble momentum P. The beta identity used here follows by the same radial and parameter integrations as Appendix M; analytic continuation supplies the dimensionally regularized relation [1]. Recursive subintegral integration has established applications to massless propagator integrals [22].

For each nonroot vertex, define

$$u_j = a_j + \sum_{h:\pi(h)=j} \sigma_h, \qquad \sigma_j = u_j + v_j - d/2. \tag{89}$$

Integrating a leaf gives $G(u_j, v_j)$ and transfers power σ_j to its parent. Repeating from the leaves to the root proves the product rule by induction. In the two-point case, set $\alpha = a_0 + \sum_{h:\pi(h)=0} \sigma_h$; its dimensionless coefficient is

$$C_{\mathcal{T}} = G(\alpha, c) \prod_{j=1}^{L-1} G(u_j, v_j). \tag{90}$$

The total physical power $A = \sum_j a_j + \sum_{j>0} v_j + c$ gives scale power $(-s - i0)^{Ld/2-A}$. Let $M_{\mathcal{T}}$ set all physical powers to one and let C_0 be its Gamma product. Restoring the original propagator signs and dividing by this master gives

$$J_{\mathcal{T}} = s^{-(A-2L)} \frac{C_{\mathcal{T}}}{C_0} M_{\mathcal{T}}. \tag{91}$$

The sign $(-1)^{A-2L}$ from the propagator convention cancels that from $(-s)^{-(A-2L)}$.

For a massive vacuum root, put $\sigma_0 = \sum_{h:\pi(h)=0} \sigma_h$. Wick rotation and the substitution $x = m^2 t$ give

$$\frac{1}{\Gamma(d/2)} \int_0^\infty \frac{x^{d/2-\sigma_0-1}}{(x+m^2)^c}\, dx = (m^2)^{d/2-\sigma_0-c} \frac{\Gamma(d/2-\sigma_0)\Gamma(\sigma_0+c-d/2)}{\Gamma(d/2)\Gamma(c)}. \tag{92}$$

Multiply the dimensionless Gamma ratio by the nonroot bubble factors to define $C^V_{\mathcal{T}}$. There are $P = 2L-1$ physical lines and $A = c + \sum_{j>0}(a_j + v_j)$. Dividing by the unit-power vacuum master $M^V_{\mathcal{T}}$ yields

$$V_{\mathcal{T}} = (-m^2)^{-(A-P)} \frac{C^V_{\mathcal{T}}}{C^V_0} M^V_{\mathcal{T}}. \tag{93}$$

In either family, a subtree with n_j vertices has $u_j = U_j - (n_j - 1)d/2$ with integer U_j. This follows from Eq. (89) by induction. Target and master Gamma arguments therefore have equal slopes in d and integer differences in their constant terms. Repeated use of $\Gamma(z+1) = z\Gamma(z)$ gives

$$\frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{k=0}^{n-1}(z+k), & n > 0, \\ 1, & n = 0, \\ [\prod_{k=1}^{-n}(z-k)]^{-1}, & n < 0. \end{cases} \tag{94}$$

Thus the target-to-master coefficients are rational in d and the one scale. Cancelling equal Gamma factors before expanding these finite products preserves the identity and avoids repeated symbolic work. This proves the reductions for the stated trees; additional coupled propagators or auxiliary numerators require another reduction.

O Direct evaluation of vacuum factors

We first identify a case that requires no master integral. If all masses vanish, there is no external momentum, and every active inverse propagator is the square of a homogeneous linear combination of loop momenta, the integrand has no dimensional scale. After Wick rotation, let R be the common radial coordinate for all loop momenta. The measure and denominators give radial power $R^{\omega-1}$ with $\omega = Ld - 2\sum_j a_j$. Split the radial integral at an arbitrary $\mu > 0$ and continue its two pieces separately:

$$\int_0^\mu R^{\omega-1}\, dR + \int_\mu^\infty R^{\omega-1}\, dR = \frac{\mu^\omega}{\omega} - \frac{\mu^\omega}{\omega} = 0. \tag{95}$$

The two pieces initially converge on opposite half-planes and their meromorphic continuations cancel. Angular singularities can be treated by analytic regularization before the same scaling argument. This is the standard scaleless-integral prescription in dimensional regularization [1]. It gives the combined integral as zero; separating ultraviolet and infrared poles requires retaining the two regulated contributions separately. The implementation checks the masses and momentum homogeneity, rejects a zero denominator with positive power, and accepts up to thirteen loops and 128 coordinates. It excludes cut denominators and symbolic-index inputs. Couplings between loop momenta are permitted for this zero-integral rule.

Consider a loop momentum k that occurs only in denominators $k^2 - x_j + i0$, where x_j is a squared mass. There is no external momentum in this component. Denominator powers can

be positive, zero, or negative. Equal masses first combine by addition of their powers. With $z = k^2$, let A_x be the resulting integer power at each distinct squared mass x. The integrand is the rational function

$$F(z) = \prod_x (z - x)^{-A_x}. \tag{96}$$

For a mass with $A_x > 0$, multiply by $(z-x)^{A_x}$ and expand the remaining analytic factor at $z = x$. Its first A_x Taylor coefficients give the principal part at that pole. Subtracting every principal part leaves a rational function without a finite pole, which is a polynomial $P(z)$. Thus

$$F(z) = P(z) + \sum_{x:A_x>0} \sum_{r=1}^{A_x} \frac{c_{xr}}{(z-x)^r}, \tag{97}$$

$$c_{xr} = \frac{1}{(A_x - r)!} \frac{d^{A_x - r}}{dz^{A_x - r}} \prod_{y \neq x} (z - y)^{-A_y} \bigg|_{z=x}. \tag{98}$$

This derivation also covers negative powers, since they supply polynomial factors in the product. For distinct rational masses, the coefficients are rational numbers. The implementation obtains them by multiplying truncated Taylor series, so it avoids general symbolic differentiation. A factor with exponent $-A_y$ has coefficient

$$[t^n](x - y + t)^{-A_y} = \binom{-A_y}{n} (x - y)^{-A_y - n}. \tag{99}$$

The binomial series proves Eq. (99); only $0 \le n < A_x$ is needed in Eq. (98). Equal masses must be combined before this expansion, which avoids division by a vanishing mass difference.

Polynomial terms in k and massless vacuum integrals have no scale and vanish in dimensional regularization [1]. For a nonzero squared mass x, define the normalized tadpole

$$T_r(x) = \int \frac{d^d k}{i\pi^{d/2}} (k^2 - x + i0)^{-r}. \tag{100}$$

To evaluate it, first take $x > 0$ and a convergent Euclidean domain. The Schwinger representation follows from the defining Gamma integral after the change of variable $u = \alpha(q^2 + x)$:

$$(q^2 + x)^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty \alpha^{r-1} e^{-\alpha(q^2+x)} \, d\alpha. \tag{101}$$

For $\alpha > 0$, let $G_1 = \int_{-\infty}^{\infty} e^{-\alpha q^2} \, dq$. Squaring this integral and using polar coordinates in two dimensions gives $G_1^2 = 2\pi \int_0^\infty r e^{-\alpha r^2} dr = \pi/\alpha$. Taking its positive square root and multiplying independent Cartesian integrals gives $\int d^d q \, e^{-\alpha q^2} = \pi^{d/2} \alpha^{-d/2}$ for positive integer d. Its analytic continuation supplies the expression used in dimensional regularization. Inserting this in Eq. (101) and setting $u = \alpha x$ gives $x^{d/2-r}\Gamma(r-d/2)/\Gamma(r)$. Wick rotation supplies $(-1)^r$ for the denominator convention in Eq. (100). Analytic continuation therefore gives

$$T_r(x) = (-1)^r \frac{\Gamma(r - d/2)}{\Gamma(r)} (x - i0)^{d/2 - r}, \qquad \frac{T_r(x)}{T_1(x)} = \frac{\prod_{j=1}^{r-1} (d/2 - j)}{(r-1)! x^{r-1}}. \tag{102}$$

The second equality follows by applying $\Gamma(z + 1) = z\Gamma(z)$ repeatedly to the first. The empty product at $r = 1$ is one. Applying Eq. (102) term by term in Eq. (97) proves both the reduction to unit tadpoles and their full analytic evaluation. The result is a meromorphic expression in d; ultraviolet or infrared poles are retained. The same tadpole value also evaluates any unit tadpole returned by the retained zero-invariant tensor formulas. A remaining master with several denominators still requires its own evaluation.

Independent components multiply, as proved in Appendix A. The direct implementation accepts one through thirteen such loop momenta, at most eight denominators per component, individual absolute powers at most eight, and total positive power after equal-mass combination at most 48 per component. Masses are nonnegative rational numbers. Denominators with nonzero target power must carry one of those individual momenta with coefficient $+1$ or -1. These tests exclude couplings between components and specify the actual scope of the rule.

The connected one-mass vacuum trees have a separate analytic evaluation. Appendix N derives their dimensionless Gamma product $C_{\mathcal{T}}^{V}$. With total physical power A, Wick rotation of all integer-power denominators gives

$$V_{\mathcal{T}} = (-1)^{A}(m^2 - i0)^{Ld/2-A} C_{\mathcal{T}}^{V}. \tag{103}$$

The exponent follows either from the final radial integral or from the mass dimension of the L measures and A denominators. For a unit-power master, $A = P = 2L - 1$. The application now returns its explicit Gamma arguments and scale exponent together with the rational reduction. This exposes the analytic master value already implied by the tree derivation and avoids a separate master evaluation for these families.

P Analytic inputs and checks

The two-mass vacuum grid contains one request at each $L = 1, \ldots, 13$. At loop order L, each independent component has squared masses $(h+2, h+7)$ with $h = 23011+L$. Its four final target power pairs are $(2,1), (3,1), (2,3), (3,-1)$, and each full target repeats the same pair across all L components. These are products of one-loop factors. The connected vacuum grid has 34 requests: chains at $L = 1, \ldots, 13$, stars at $L = 3, \ldots, 13$, and binary trees at $L = 4, \ldots, 13$. Their parent maps are $\pi(j) = j - 1$, $\pi(j) = 0$, and $\pi(j) = \lfloor (j-1)/2 \rfloor$, respectively. These starting orders omit coincident small-loop routings. The root squared mass remains symbolic. Each final request has root power two, root power three, root and final edge powers two, and root power four, with every other physical power one. The one-loop request instead uses powers two through five. The unit-power integral remains the master. The profiles are $23111 + L$, $23211 + L$, and $23311 + L$ for the three shapes and fix any auxiliary construction choices. These requests test preservation of the connected identity and the earlier symbolic single-loop dispatch.

The repeated-mass coverage grid uses four denominators per component with squared masses $(h+2, h+7, h+2, 0)$, where $h = 23511+L$. The four power assignments are $(1,1,2,0)$, $(2,1,1,1)$, $(1,2,0,-2)$, and $(0,0,0,2)$. This tests repeated masses, a massless pole, a polynomial numerator, and a scaleless target. It is checked through all thirteen loop orders, with no speed claim against an unfinished or unsupported native calculation.

The scaleless coverage grid has one family for each $L = 1, \ldots, 13$, with inverse propagators ℓ_i^2 and $(\ell_i - \ell_j)^2$ for all $i < j$. It therefore contains $L(L+1)/2$ coordinates and coupled loop momenta for $L > 1$. The three targets set all powers to one, raise the first power to two, or change the first power to minus one. Equation (95) supplies the reference zero. Native one- and two-loop checks test the same rule separately.

The three retained zero-invariant checks use the unequal, equal, and mixed mass choices of Eq. (23), with profiles $23211, 23212, 23213$, numerator rank two, and two excess positive powers. The second excess power is placed on the first physical denominator. The three ordinary tensor checks use Eq. (24) with $h = 23311 + P$ for $P = 2, 3, 4$. Their numerator rank and excess positive power are both two, with one of the excess powers placed on the first physical denominator.

The new native reconstruction inputs are specified in Appendix Q.

An independent symbolic partial-fraction calculation checks the vacuum grid. Random rational-function checks verify that subtracting the proposed poles leaves only a polynomial. Convergent Euclidean radial quadrature checks the tadpole normalization, and a native IBP compari-

son checks its raised-power recurrence. Native box and ordinary tensor coefficients are checked against independently prepared Kira reductions. The connected tree comparison checks preservation against the preceding exact rule; its mathematical justification is the separate derivation in Appendix N.

Q Native reconstruction inputs

The planar and nonplanar boxes use Eqs. (25) and (26), respectively. The high-rank targets are $(1,1,1,1,1,1,1,-u,-v)$ with $u,v \geq 0$ and $1 \leq u+v \leq 4$. Each graph contributes fourteen training, six validation, and six reserved high-rank batches. The complete target sets differ between splits and exclude the preceding comparison batches. Individual target vectors can occur in more than one batch.

For batch index $i = 0,\ldots,25$, define $j = i \bmod 6$ and $q = \lfloor i/6 \rfloor$. Let

$$\begin{aligned}(A_0,\ldots,A_5) &= (3,13,103,1009,10007,100003),\\ (B_0,\ldots,B_5) &= (7,19,107,1013,10009,100019).\end{aligned} \tag{104}$$

We set $s = -(A_j + 2q)$ and $t = -(B_j + 4q)$ in both graphs, using each graph's scalar-product convention. Indices 0–13 train the model, 14–19 select it, and 20–25 test it. These equations define the chosen input points. In the planar graph the fourth momentum is $-p_1 - p_2 - p_3$, and in the nonplanar graph it is $p_1 + p_2 - p_3$. Expanding their squares with the listed scalar products gives zero in both cases. The test batches are specified in Table 15.

Table 15: Reserved high-rank target batches. The same (u,v) sets and invariants enter both box routings. Every row defines a complete request for each graph.

i	s	t	Numerator powers (u,v)
20	-109	-119	$(2,0),(0,2),(0,1)$
21	-1015	-1025	$(2,2),(0,4)$
22	-10013	-10021	$(3,1),(3,0),(2,1),(1,3),(0,1)$
23	-100009	-100031	$(3,1),(2,2),(2,1),(2,0),(1,2),(0,4)$
24	-11	-23	$(2,2),(2,1),(1,2),(1,1),(0,2),(0,1)$
25	-21	-35	$(2,1),(1,3),(1,2),(1,0),(0,2),(0,1)$

For the lower-rank box requests, the sector corner is $(1,1,1,1,1,1,1,0,0)$. A numerator request uses the supplied first-rank auxiliary targets, and a raised-power request increases one physical index. Table 16 lists their precise target vectors. Their profiles are $h = 27701, 27713$ for training, 27733 for validation, and 27739 for testing, with the low-rank kinematics of Appendix D.

The sunrise family has loop momenta (ℓ_1,ℓ_2) and one external momentum p. Its inverse-propagator momenta and squared masses are

$$(Q_1,\ldots,Q_5) = (\ell_1,\ell_2,p-\ell_1-\ell_2,p-\ell_1,p-\ell_2), \qquad (m_1^2,\ldots,m_5^2) = (3,7,13,0,0). \tag{105}$$

The first three coordinates are physical denominators and the last two are auxiliary. The three-loop vacuum uses

$$\begin{aligned}(Q_1,\ldots,Q_6) &= (\ell_1,\ell_2,\ell_3,\ell_1-\ell_2,\ell_2-\ell_3,\ell_1-\ell_3),\\ (m_1^2,\ldots,m_6^2) &= (1,1,1,13,13,2h+53).\end{aligned} \tag{106}$$

All six coordinates are physical. Both structures use the same profile split as the low-rank boxes. For the sunrise, $p^2 = -1009 - 12h$. These assignments specify test families, rather than universal relations between masses and invariants. In every native request the kinematic quantities are

Table 16: Lower-rank reserved targets. Training and validation use the same index patterns at different physical points. A corner request contains the all-unit physical sector.

Structure and request	Target indices
Planar box, corner	$(1,1,1,1,1,1,1,0,0)$
Planar box, numerator	$(1,1,1,1,1,1,1,0,-1), (1,1,1,1,1,1,1,-1,0)$
Planar box, raised power	$(1,1,1,1,1,2,1,0,0), (1,1,1,1,1,1,2,0,0), (2,1,1,1,1,1,1,0,0)$
Nonplanar box, corner	$(1,1,1,1,1,1,1,0,0)$
Nonplanar box, numerator	$(1,1,1,1,1,1,1,0,-1), (1,1,1,1,1,1,1,-1,0)$
Nonplanar box, raised power	$(1,1,1,1,1,2,1,0,0), (1,1,1,1,1,1,2,0,0), (2,1,1,1,1,1,1,0,0)$
Sunrise, corner	$(1,1,1,0,0)$
Sunrise, raised power	$(1,2,1,0,0), (1,1,2,0,0), (2,1,1,0,0)$
Vacuum, corner	$(1,1,1,1,1,1)$
Vacuum, raised power	$(1,2,1,1,1,1), (1,1,2,1,1,1), (1,1,1,2,1,1)$

fixed rational numbers and d remains symbolic, so the coefficients belong to $\mathbb{Q}(d)$. This test does not reconstruct general symbolic scattering invariants.

The protected native bases consist of the integral indices below. Each listed vector $\boldsymbol{b}$ denotes $M_{\boldsymbol{b}} = I_{\boldsymbol{b}}$ in its corresponding family. The independently prepared symbolic and reconstructed calculations use the same basis for each request. The lists specify the comparison basis; they do not assert a proof of global master independence from a finite seed system.

Planar box:

$$\{(0,0,1,1,0,0,1,0,0), (0,1,0,0,0,1,1,0,0),\\ (0,1,0,1,1,0,1,0,0), (0,1,1,1,0,1,1,0,0),\\ (1,0,1,1,1,0,0,0,0), (1,1,1,0,0,1,1,0,0),\\ (1,1,1,1,1,1,1,-1,0), (1,1,1,1,1,1,1,0,0)\}.$$

Nonplanar box:

$$\{(0,0,1,1,1,1,0,0,0), (0,1,0,0,1,1,0,0,0),\\ (0,1,1,1,1,1,1,0,0), (1,0,0,1,0,1,0,0,0),\\ (1,0,1,1,1,1,0,0,0), (1,1,0,0,0,0,1,0,0),\\ (1,1,0,1,0,1,1,0,0), (1,1,0,1,1,1,0,0,0),\\ (1,1,1,0,1,0,1,0,0), (1,1,1,1,0,0,1,0,0),\\ (1,1,1,1,1,1,1,-1,0), (1,1,1,1,1,1,1,0,0)\}.$$

Sunrise, basis 1:

$$\{(0,1,1,0,0), (1,0,1,0,0),\\ (1,1,0,0,0), (1,1,1,-1,0),\\ (1,1,1,0,-1), (1,1,1,0,0),\\ (1,1,2,0,0)\}.$$

Sunrise, basis 2:

$$\{(1,1,1,0,0)\}.$$

Three-loop vacuum, basis 1:

$$\{(0,1,1,1,0,1),(0,1,1,1,0,2),\\ (0,1,1,2,0,1),(1,0,0,1,1,1),\\ (1,0,1,1,0,1),(1,0,1,1,1,0),\\ (1,0,1,1,1,1),(1,0,1,1,2,0),\\ (1,1,0,1,0,1),(1,1,0,1,1,0),\\ (1,1,0,1,1,1),(1,1,1,0,0,1),\\ (1,1,1,1,0,0),(1,1,1,1,0,1),\\ (1,1,1,1,1,0),(1,1,1,1,1,1)\}.$$

Three-loop vacuum, basis 2:

$$\{(1,1,1,1,1,1)\}.$$

R Coefficient reconstruction

Kira and FireFly provide the finite-field algorithms used here [9, 10, 11]. To explain the relation to an exact rational coefficient, suppose

$$r(d) = P(d)/Q(d), \qquad P(d) = \sum_{j=0}^{n} p_j d^j, \quad Q(d) = \sum_{j=0}^{m} q_j d^j. \tag{107}$$

One nonzero denominator coefficient fixes the overall normalization. At a nonsingular point $d = z$ modulo a prime, a numerical reduction returns $r(z)$. Multiplying by the denominator gives the linear constraint

$$\sum_{j=0}^{n} p_j z^j - r(z) \sum_{j=0}^{m} q_j z^j = 0 \pmod{p}. \tag{108}$$

This follows directly from Eq. (107). With correct degree bounds and sufficiently many independent nonsingular samples, the constraints determine the normalized coefficients over the field. Combining suitable primes and reconstructing rational numbers then supplies rational coefficients [10, 11]. FireFly determines the necessary information adaptively; our model chooses the native procedure and probe batch size.

The field operations are exact, while reconstruction termination relies on its degree discovery and checks. Our evaluation therefore also subtracts every reconstructed coefficient from a separately computed symbolic reference and requires the rational difference to vanish. This checks the actual tested answer. Neither a model score nor a successful reduction to the supplied basis alone proves every coefficient for an arbitrary unseen request.

S Exact reduction of a one-scale massless triangle

The on-shell triangle has a Gamma-function representation [16]. We derive its coefficient in our normalization to specify the signs and the master basis used by the implementation. Let

$$T_{abc}(Q,d) = \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{[k^2+i0]^a[(k-p_1)^2+i0]^b[(k-p_1-p_2)^2+i0]^c}, \tag{109}$$

where $p_1^2 = p_2^2 = 0$, $Q = -(p_1+p_2)^2 > 0$, and a, b, c are propagator powers. Write $A = a+b+c$ and $h = d/2$. For initially positive powers in a convergence region, Feynman parametrization and the shifted loop integral give

$$T_{abc} = \frac{(-1)^A \Gamma(A-h)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_{x_j \geq 0} d^3x\, \delta(1-x_1-x_2-x_3) x_1^{a-1} x_2^{b-1} x_3^{c-1} (Q x_1 x_3)^{h-A}. \tag{110}$$

The only nonzero squared momentum difference is between the first and third propagators, which gives the factor Qx_1x_3. The Dirichlet integral follows by writing three Gamma integrals and changing their variables to a common scale and three fractions. Explicitly, the substitution $t_j = ux_j$ in $\prod_j \int_0^\infty dt_j\, t_j^{z_j-1} e^{-t_j}$ gives

$$\int d^3x\, \delta(1-\textstyle\sum_j x_j) \prod_j x_j^{z_j-1} = \dfrac{\prod_j \Gamma(z_j)}{\Gamma(\sum_j z_j)}. \tag{111}$$

Applying this identity with $z_1 = h-b-c$, $z_2 = b$, and $z_3 = h-a-b$ yields

$$T_{abc} = (-1)^A Q^{h-A} \frac{\Gamma(A-h)\Gamma(h-b-c)\Gamma(h-a-b)}{\Gamma(a)\Gamma(c)\Gamma(d-A)}. \tag{112}$$

Dimensional continuation defines the expression outside the initial convergence region. The limit $b \to 0$ gives the two-point integral obtained by removing the middle propagator. Its unit-power value is

$$B(Q,d) = T_{101} = Q^{h-2} \frac{\Gamma(2-h)\Gamma(h-1)^2}{\Gamma(d-2)}. \tag{113}$$

For positive integers a, c and a nonnegative integer b, division of Eq. (112) by Eq. (113) gives a rational coefficient. To evaluate it with finite products, define

$$(z)_n = \frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{j=0}^{n-1}(z+j), & n \geq 0, \\ \dfrac{1}{\prod_{j=1}^{-n}(z-j)}, & n < 0. \end{cases} \tag{114}$$

The empty product equals one. The identity follows by repeated application of $\Gamma(z+1) = z\Gamma(z)$ and gives

$$T_{abc} = R_{abc}(Q,d) B(Q,d), \tag{115}$$

$$R_{abc} = \frac{(-1)^A Q^{2-A}}{(a-1)!(c-1)!} (2-h)_{A-2} (h-1)_{1-a-b} (h-1)_{1-b-c} (d-A)_{A-2}. \tag{116}$$

For example, $R_{111} = 2(d-3)/[Q(d-4)]$. Increasing a by one gives the recurrence

$$R_{a+1,b,c} = -\frac{(A-h)(d-A-1)}{Qa(h-a-b-1)} R_{abc}, \tag{117}$$

which supplies a separate exact check on the finite-product calculation. If $a = 0$ or $c = 0$, the remaining massless denominators have no nonzero external invariant. Those sectors vanish in dimensional regularization [1].

The implementation checks all masses and pairwise momentum differences before using this formula. It accepts rational $Q > 0$, powers from zero through four, and $A \leq 8$. It rejects cut propagators, additional occupied lines, and negative indices. For independent loop components with no occupied cross-loop denominator, the product measure then gives

$$\prod_{j=1}^{L} T_{a_j b_j c_j}(Q_j, d) = \left[\prod_{j=1}^{L} R_{a_j b_j c_j}(Q_j, d) \right] \prod_{j=1}^{L} B(Q_j, d). \tag{118}$$

The proof follows by factoring the integrand and measure in a common convergence region and then continuing the result analytically. If the requested integer powers have no common convergence region in d, we first allow the propagator powers to vary independently. Equation (112) gives their meromorphic continuation, which defines the product at the requested powers and dimension. This extension covers products through thirteen loops in the implementation. The implementation retains this product of component coefficients instead of expanding it into a larger polynomial. It supplies no formula for a general connected thirteen-loop scattering graph.

T Inputs for the final triangle and native comparison

The four native propagator structures are those in Appendix Q. For the new planar and nonplanar box requests, the invariants are $s = -(137+6i)$ and $t = -(167+4i)$, with the incoming/outgoing momentum conventions stated there. Training uses $i = 2, \ldots, 13$, validation uses $i = 14, \ldots, 18$, and reserved tests use $i = 19, \ldots, 23$. Each box target has powers $(1, 1, 1, 1, 1, 1, 1, -a, -b)$. The reserved pairs are listed below; both box structures use the same pairs at each i.

i	Pairs (a, b) in the target request
19	$(3,0), (2,2), (0,2)$
20	$(4,0), (3,1)$
21	$(4,0), (1,2), (0,2), (0,1)$
22	$(3,1), (2,1), (0,2)$
23	$(2,1), (2,0), (1,0), (0,4), (0,2), (0,1)$

The sunrise masses remain $(3, 7, 13)$ and its external invariant is $p^2 = -386221 - 12i$. The three-loop vacuum masses remain $(1, 1, 1, 13, 13, 64255 + 2i)$. These families use the same index splits. Even i requests the sector corner, and odd i requests the raised-power targets defined for that structure. Full physical inputs differ between splits. Graph structures and individual index vectors recur; box target batches are distinct across the new splits.

For a triangle product with L loops, every factor has $Q = 101 + 2L$. We choose factor powers from $(1,1,1)$, $(2,1,1)$, $(1,2,1)$, $(1,1,2)$, $(2,2,1)$, $(1,0,1)$, and $(2,0,2)$. We form six products per loop order and remove duplicate full targets. Cross-loop auxiliary powers are zero. Seven additional single-loop requests use $Q = 211 + 2j$, $j = 1, \ldots, 7$, and the targets $(1,1,1)$, $(1 + (j \bmod 3), 1 + (j \bmod 2), 1)$, $(1,0,1)$, $(2,0,1)$, and $(0,1,1)$, again with duplicate targets removed.

The preceding triangle basis uses unit-power triangle masters. For verification, each such master is replaced by $2(d-3)B/[Q(d-4)]$, and independent components multiply. Kira provides a separate bubble-basis reference for each new single-loop request. The higher-loop comparison combines this exact basis conversion with the product derivation and independent Gamma-function checks.

U Details of the final incremental comparison

We use complete reduction requests to test the method, since fewer seeds or a more accurate label prediction does not determine the elapsed time. This section states what the requests contain, which calculations serve as controls, and what enters each timer. All numerical results below are measurements from this study; external references support the established methods, while the appendices derive the identities and estimators used here.

U.1 The final collection

The final collection has 93 requests. It combines 40 new requests with 53 previously checked analytic requests that test whether the existing exact methods still work. Table 17 identifies the physical structures. The exact propagators, master lists, target vectors, and kinematic assignments are given in Appendices D, P, Q, and T.

The names planar and nonplanar refer to the line-connection structures of the two box families. Their equations in Appendix D specify the exact meaning used here. Both structures appear in training, validation, and testing. The new native tests change the physical points and target batches, so they measure transfer within known structures. They do not hold out every graph structure. Individual index patterns can also recur between splits.

Table 17: The 93 requests in the final comparison. A request can contain several targets. Native inputs use rational masses and external invariants with symbolic dimension d.

Structure	Loops	Requests	Reduction and scope
Planar and nonplanar boxes	2	$5+5$	Four external legs; native IBP reduction
Massive sunrise	2	5	Two-point family with three massive physical lines; native IBP reduction
Massive vacuum family	3	5	Six coupled physical denominators; native IBP reduction
Massless triangles and independent products	1–13	20	One nonzero invariant per triangle; direct formula
Independent vacuum products	1–13	13	Two squared masses per component; partial fractions
Connected one-mass vacuum trees	1–13	34	Chain, star, and binary-tree connections; successive integrations
Zero-invariant and ordinary one-loop numerator integrals	1	$3+3$	Tensor recurrences and the endpoint formula
Total		93	20 native and 73 analytic requests

For the new box inputs, training uses indices $i = 2, \ldots, 13$, validation uses $i = 14, \ldots, 18$, and testing uses $i = 19, \ldots, 23$, with the physical assignments in Appendix T. Targets include numerator powers and raised physical denominator powers. The independent triangle and vacuum products include every loop order from one through thirteen. Connected tree tests cover the same loop range under their specified one-scale assumptions.

U.2 Controls and time measurements

The final experiment compares three applications on exactly the same 93 requests. The *preceding application* is the published method before the direct triangle addition and its new native-setting selection. The *intermediate application* adds those changes with the original runtime monitor. The *final application* also corrects the monitor's handling of temporary entries that disappear while a reduction runs. The inherited seed model is shared by all three. The added arithmetic selector and equation filter are disabled by validation in the final method.

The native implementation uses Kira 3.1 with the retained project reduction choices; its reconstruction-enabled build includes FireFly. Each application processes the collection in one call. We use one worker on the same Intel Gold compute node, one computational thread per application, and three repetitions. We rotate method order and reverse request order in the middle repetition. The measured time includes loading, method selection, algebra, reading outputs, cleanup, and actual recovery. Preparation of independent reference reductions and the external exact coefficient comparisons lie outside every application timer. The comparison therefore measures the reduction application, with an additional untimed correctness audit.

We report the median of the three complete-collection times and their observed minimum and maximum. These three-observation ranges are not confidence intervals. For diagnostic results within a physical group, we sum the per-request median times recorded inside the shared call. These component sums omit some shared overhead and are not the complete-collection estimator. Appendix E states both definitions.

The native references are independently prepared symbolic reductions in the prescribed bases. Independent symbolic partial fractions check the vacuum products. Tree checks compare the retained recurrence with the preceding exact result and its analytic derivation. Triangle checks use exact conversion between the preceding triangle basis and the new bubble basis, separate one-loop Kira references, and Gamma-function identities. Analytic products are justified by the

factorization proof. The notes state the individual check counts and preserve the unsuccessful trials.

U.3 Separate comparisons needed to identify the cause of a gain

The final baseline already contains exact formulas, learned seeds, and earlier Kira improvements. Its time ratio therefore measures the latest combined change; it does not measure a speedup relative to unmodified Kira. We retain three other comparisons to identify what the individual methods contribute.

First, a fresh-call comparison measures 40 new requests separately, including a new process for each request. Second, an earlier 75-request comparison isolates the addition of direct vacuum formulas and validated native reconstruction settings on one matched workload. Third, model-on/model-off comparisons hold the other procedures fixed. An *ablation* is such a comparison with one component removed. We use it to distinguish learned savings from analytic formulas, fixed tuning, and shared work.

The older fresh-call comparisons use sums of per-request median times. Where stated, their uncertainty intervals resample paired requests 5,000 times and report the central 95% of the resulting ratios, following the bootstrap method [19]. This measures variation across the sampled requests. It gives no confidence statement for all possible integral families. We report each workload and control beside its result and do not multiply ratios from different tests.

V Final incremental results

The combined method reduces the time for the final collection, but the contribution depends on the physical structure and on how the requests are processed. We first report the complete comparison, then the individual families and learned components.

V.1 Complete time for the final 93 requests

The final application takes a median of 41.137 seconds, compared with 45.091 seconds for the preceding application. The ratio is 0.912, which corresponds to 8.8% less time. Table 18 gives all three measured procedures, and Figure 5 shows each repetition. All 837 application results, from 93 requests, three methods, and three repetitions, pass the 13,401 exact coefficient comparisons. No internal native attempt fails in this final serial comparison.

Table 18: Complete-collection elapsed time for 93 matched requests. The ranges contain the three observed times; they are not confidence intervals. All methods process the collection in one call.

Application	Median (s)	Observed range (s)
Preceding	45.091	44.783–45.817
Intermediate	42.932	40.351–44.589
Final	41.137	40.723–41.227

The intermediate-to-final difference is smaller than the preceding-to-final difference and varies with the execution conditions. The monitor correction removes a reproduced interruption mechanism, but these timings alone do not assign all of their difference to that correction. The scientific result is the measured total change of the combined procedure.

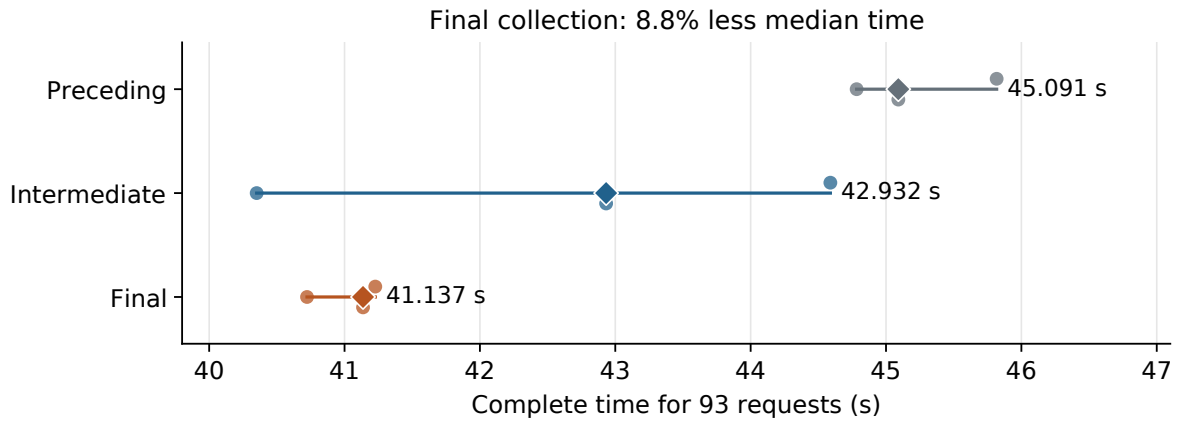


Figure 5: The final method reduces complete-collection time in each recorded repetition. Points are the three observations for each method; diamonds are their medians. The elapsed-time axis is restricted to the observed range so that the comparison remains readable. All points include shared setup and recovery costs.

V.2 Which integral structures become faster

Table 19 shows the native groups within the final shared-call experiment. Planar and nonplanar boxes become faster, while the sunrise group becomes slower and the three-loop vacuum group changes little. Thus an aggregate improvement does not imply an improvement for every family. The component times also show that the remaining native algebra dominates this particular collection.

Table 19: Native component times inside the final shared-call comparison. Each entry is a sum of per-request medians for five requests. Ratios divide final time by preceding time; shared overhead is excluded from these component sums.

Structure	Preceding (s)	Final (s)	Ratio
Planar box	8.948	7.191	0.804
Nonplanar box	14.517	11.137	0.767
Massive sunrise	7.509	8.407	1.120
Three-loop vacuum	12.971	12.988	1.001

The separate fresh-call experiment asks a different question: how much time each new request needs when it starts its own application. Across its 40 requests, summed median time falls from 57.766 to 50.848 seconds, a ratio of 0.880. The paired request-resampling interval is $[0.744, 0.992]$. The 20 native requests alone give 0.887 with interval $[0.743, 1.010]$, which includes equal time. These results refer to the direct triangle and native-setting changes before the final monitor correction.

Figure 6 shows the fresh-call change by physical structure and the triangle-product points at each loop order. The triangle group gives a ratio of 0.816, but the seven-loop product gives 1.077 and is slower. Higher-loop products test the factorization identity under the specified powers and invariants. They supply no timing result for a general connected scattering graph at the same loop order.

V.3 How much of the gain comes from machine learning

Earlier controlled tests resolve small seed-selection and equation-selection gains on their stated workloads (Table 20). These results motivated the learned interventions. The row model took

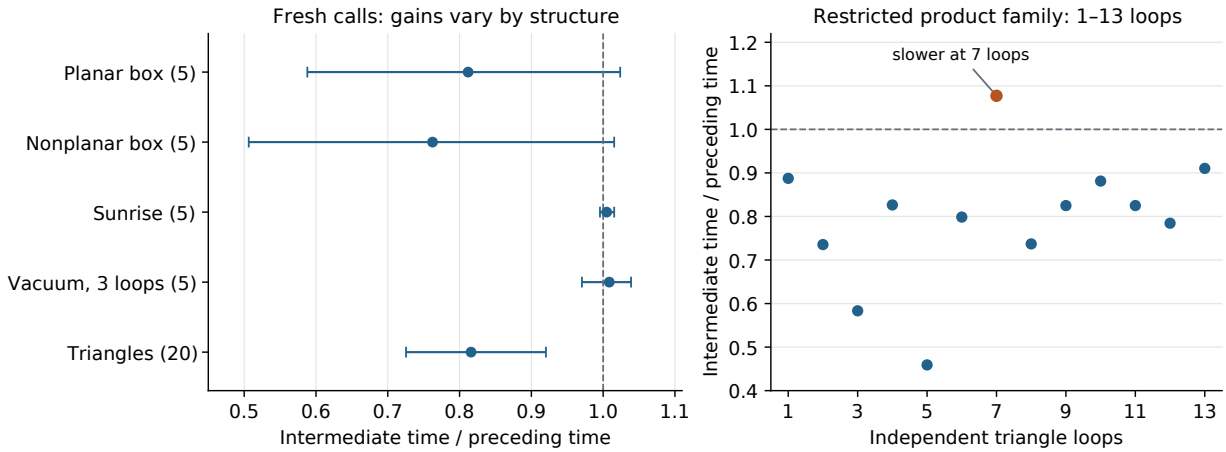


Figure 6: The fresh-call gain depends on the input. Left: ratio of summed median times for each group; each entry uses the same requests in both methods, with request counts in parentheses. Horizontal bars are central 95% paired-request bootstrap intervals. Right: triangle-product ratios at loop orders one through thirteen. There are eight requests at one loop and one at each higher loop order; the right panel therefore shows point estimates without intervals. Values below the dashed line mean less time. These plots show the intermediate method against the preceding method, and use a different timing protocol from Figure 5.

more time, and a later compiled row policy had an interval that included equal time. We therefore retain Kira's native row choices.

Table 20: Separate learned-component comparisons from this study. Ratios below one mean less time. The workloads and controls differ, so the ratios cannot be combined into one speed factor. Intervals retain the original study's resampling units, specified below.

Learned component	Control and workload	Ratio	Interval
Earlier seed model	Fixed seed region; 32 box batches	0.963	$[0.937, 0.984]$
Earlier equation model	Native filter; 24 new-index batches	0.933	$[0.928, 0.940]$
Original row model	Native row choices; 84 trials	1.305	—
Compiled row-cost policy	Native choices; geometric-mean estimator	0.982	$[0.959, 1.003]$
Inherited learned choices with reconstruction	Fixed choices, same arithmetic; 22 native requests	0.983	$[0.951, 1.003]$

The seed and reconstruction-era intervals resample requests; the equation interval resamples paired repetitions on fixed requests. The equation timer includes its coefficient checks and excludes initial imports, whereas the final collection has the timing boundary in Appendix U. The compiled-row interval retains its original grouped estimator. These differences are another reason to keep the comparisons separate.

The last row tests the inherited learned choices after the arithmetic changes. Its fixed and learned times are 42.175 and 41.448 seconds, respectively, but the interval includes equal time. It therefore does not establish a reliable additional aggregate gain from those learned choices on that sample. Both procedures complete all requests without an unsuccessful internal attempt.

The latest arithmetic selector also fails its validation condition. Its best fitted choice has 46.495 seconds of validation cost, compared with 46.438 seconds for the structure-dependent fixed settings. The required three-percent gain is absent, so the added selector is disabled. Similarly, the nonplanar equation filter takes 16.807 seconds against 9.824 seconds without the filter. It skips 17,808 of 53,430 candidate equation constructions, but all ten filtered attempts fail to reach the prescribed master basis and need recovery. This test demonstrates why equation counts alone cannot establish a time gain.

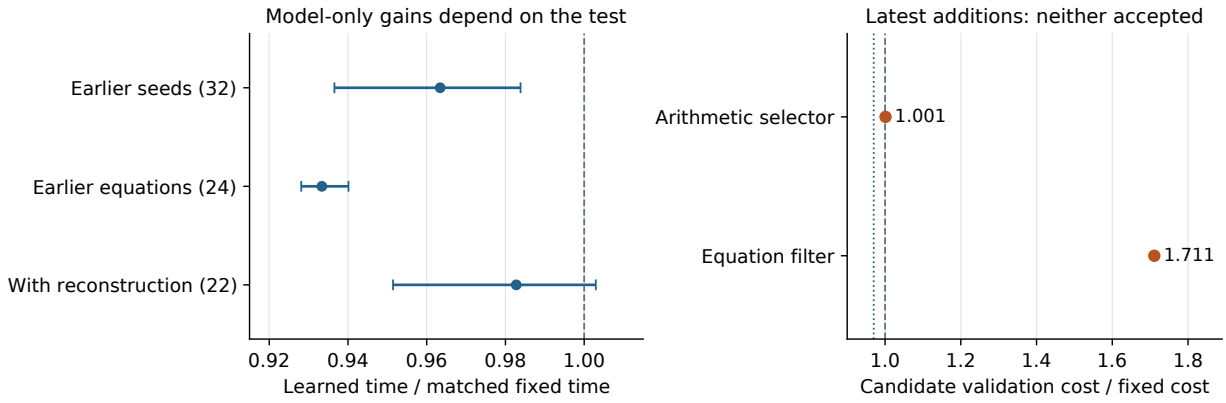


Figure 7: Learned improvements are specific to a tested setting. Left: three model-only time ratios with their recorded resampling intervals and request counts. The seed and reconstruction intervals resample requests; the equation interval resamples paired repetitions. The reconstruction-era interval includes equal time. Right: latest validation costs, normalized to the matched fixed control. The additional arithmetic selector and equation filter remain disabled. The dashed line marks equal cost, and the dotted line marks the arithmetic selector's 0.97 acceptance threshold.

V.4 Earlier combined gains and unsuccessful comparisons

The earlier 75-request experiment measures the effect of adding validated coefficient reconstruction to an application that already has direct vacuum formulas. Summed median time changes from 79.624 to 49.852 seconds, or 37.4% less time. The directly matched comparison across both the vacuum and reconstruction changes gives 81.827 to 49.852 seconds, or 39.1% less time. The latter is a cumulative result on one workload. Its control is an earlier application, not unmodified Kira, and its percentage is separate from the final 93-request result.

An initial parallel version of the final comparison stopped before all three repetitions finished. Of 651 recorded request results, 650 passed and one preceding-method vacuum result failed the prescribed-basis check. After an interrupted attempt, recovery returned 20 masters while the reference used 16. No conversion between those bases was performed, so the test did not establish whether the returned expressions were equivalent. We preserve this failed comparison and exclude its incomplete repetition from the timing estimator. The serial experiment of Table 18 is complete; it does not erase the earlier failure. The notes retain the full interrupted experiment, older timing protocols, and model-training results.

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